

Gender Quota Laws and Women in Cabinets¹

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Abstract

Do legislative gender quotas increase women's presence in cabinets? Women remain underrepresented in political leadership worldwide. As a remedy, over 80 countries have adopted gender quotas, requiring parties to nominate or elect a minimum share of women. But can quotas have effects beyond the positions they directly target? We argue that quotas increase the presence of women in executive cabinets by expanding the pool of experienced female legislators. Using a global dataset (168 countries) from 1990 to 2021, we find gender quotas increase the share of women ministers by 15 percent relative to the average baseline—including increases in both high- and low-prestige portfolios. Consistent with a supply-side mechanism, effects are largest in parliamentary democracies—where ministers are often selected from parliament—and in countries that experienced the greatest increases in women's legislative representation post-quota. The findings suggest quotas can generate meaningful spillover effects at the highest levels of government.

Supplementary material for this article is available in the appendix in the online edition.

Replication files are available in the JOP Dataverse (<https://dataverse.harvard.edu/dataverse/jop>). The empirical analysis has been successfully replicated by the *JOP* replication analyst.

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In 2011, liberal party politician Annemie Turtelboom became the first woman Minister of Justice in Belgium. She attributed her political success to Belgium's gender quota law, noting: "I am the product of quotas, and I feel good about that. Without quotas in politics, women would never have been able to break through certain boundaries."¹ This example suggests that quota laws, which mandate women's inclusion on candidate slates or in office, might chip away at some of the most intractable gender gaps in access to top leadership roles. Yet, legal quotas are controversial policies which can cause backlash against women. This raises an open question that has important implications for women's representation and the quality of democracies: Can gender quotas work beyond the positions to which they apply?

By 2021, roughly half of countries worldwide had adopted a legal gender quota for the national parliament.² As countries across the globe adopted gender quotas, many parliaments went from having only superficial numbers of women to women holding a substantial share of seats overnight (Hughes et al. 2019; Paxton and Hughes 2015). Although advocates of quotas laud their success in "fast-tracking" women into some of the most powerful decision-making bodies in the world (Tripp and Kang 2008), it is unclear whether quotas are sufficient to transform politics (Murray 2013). When designed properly, quotas are effective at increasing women's numeric representation (Schwindt-Bayer 2009; Aldrich and Daniel 2024). But, single country studies show mixed evidence about whether quotas have 'spillover' effects beyond the positions they directly target (O'Brien and Rickne 2016; Kerevel 2019). We investigate whether legal quotas—constitutional or electoral mandates that apply to all political parties—can transform political power by increasing women's access to executive cabinets.

¹Door, Joel de Ceular and Walter Pauli. "Als Minister Moet Je Soms Drie Dagen Geen Krant Lezen"; Minister van Justitie Annemie Turtelboom (Open VLD)." 16 October 2013. *Knack Magazine*. Translation by Google Translate.

²As of 2021, 84 of 168 countries (50%) in our sample had a national legislative gender quota (reserved seats or candidate quotas, not voluntary party quotas; Hughes et al. 2019).

We argue that quotas increase the supply of women ministers by creating a larger pipeline of women with the experience often necessary for selection (Aldrich and Daniel 2025; Annesley, Beckwith and Franceschet 2019). Quotas fast-track women's entry into parliaments, and the share of women in parliaments tracks closely with the share of women in governments (Krook and O'Brien 2012; Stockemer 2017). In addition, we investigate potential demand-side effects. By elevating the issue of gender equality in politics to the national stage, quotas may raise expectations that parliaments *and governments* should include more women. As these laws are implemented, both the public and party elites are exposed to new conceptions of what politicians can look like (Barnes and Holman 2020; Taylor-Robinson and Geva 2023).

We test this argument using a global sample of 168 countries from 1990 to 2021, applying a range of methods (descriptive trend analysis, difference-in-differences, and causal mediation). In line with our main argument, we find that legal quotas are followed by a 2.1–2.6 percentage-point rise in women's cabinet share—a 15 percent gain over the baseline. Our analysis reports strong, consistent evidence for supply-side arguments. Effects are strongest in parliamentary democracies, where ministers are typically recruited from parliament, and in countries that experience a large post-quota shock of women legislators.

Disaggregating by ministerial prestige shows women's gains are broad-based: the largest (though statistically imprecise) increases appear in low- and mid-prestige portfolios, while a smaller but statistically significant rise occurs in high-prestige posts. Quotas therefore open some doors to the most influential offices rather than confining women to the periphery. Importantly, we detect no reduction in men's cabinet seats, implying that governments often expand portfolios rather than replace male ministers. Overall, quotas generate meaningful spillovers into the executive without eroding entrenched male dominance at the top.

Women in Parliaments, Quotas, and Executive Power

The executive branch was once considered the last bastion of male political power. Today it is “internationally understood that cabinets should contain women,” and women are present in almost every cabinet across the globe (Paxton, Hughes and Barnes 2020, 171). Women’s inclusion in the cabinet is important because the executive branch is both the face of government (Annesley and Franceschet 2015, 613) and it wields the most policy making authority (Atchison and Down 2019; Homola 2022). Indeed women’s presence in the executive branch, represents both the “most significant venue for achieving the substantive representation of women” (Annesley and Gains 2010, 909), and the potential to shape citizens’ evaluations of both government and women’s ability to lead (Alexander and Jalalzai 2020).

A large body of research examines women’s access to the executive branch (Bauer and Tremblay 2011; Bauer and Darkwah 2022; Field 2021). Women’s cabinet representation is best explained by political factors, rather than “slower-moving processes of women’s social and economic empowerment” (Krook and O’Brien 2012, 850). Though some autocrats include women in their cabinets (Kroeger and Kang 2022), democratic leaders are more likely to do so (Nyrup, Yamagishi and Bramwell 2023). Left-leaning governments tend to appoint more women to cabinets (O’Brien et al. 2015; Goddard 2021), though this pattern may be dissipating over time (Stockemer and Sundström 2018). Selectors also have strategic motivations for appointing women, including electoral competition (Escobar-Lemmon and Taylor-Robinson 2016), patronage-based alliances (Arriola and Johnson 2014), and signaling their administrative priorities to domestic and international audiences (Bego 2014; Kroeger and Kang 2022).

One of the strongest and most consistent findings is that the share of women in parliament correlates with the share of women in cabinets. Analyzing 194 countries (1965-2014), Stockemer (2017) observes a strong positive relationship between the two. Global analyses from a single point in time likewise uncover the same trend (Krook and O’Brien

2012). Similar results appear in studies focused on Western Europe (Claveria 2014; O'Brien et al. 2015), Africa (Arriola and Johnson 2014; Kroeger and Kang 2022), and Latin America (Escobar-Lemmon and Taylor-Robinson 2016). This link tends to be especially strong in parliamentary systems, but some find it holds in presidential systems as well (Stockemer 2017). The relationship is so robust that Krook & O'Brien (2012, 853) conclude:

“Perhaps the most striking finding was that in a majority of cases women’s cabinet appointments were explained by the presence of more women among political elites — something that can be promoted through concrete political strategies like gender quotas...”

Despite this, we know little about how quotas affect women’s appointments to the executive branch. In a study of Mexico, Kerevel (2019) traces the careers of individual legislators, showing that women elected via quotas are less likely than men to be appointed to the cabinet after leaving office. Likewise, examining advanced industrial democracies, Claveria finds PMs from parties with voluntary quotas appoint more women than PMs from parties without quotas—reasoning that “parties using quotas are committed to equality values” (Claveria 2014, 1162). Similarly, a global study (154 countries, 1966-2015) by Lee (2024) finds a subset of “effective quotas”—those meeting a *de facto* threshold of 10% and backed by sanctions or placement mandates (Hughes et al., 2019)—increase the long-term share of women in cabinet, with the strongest effects emerging 15 to 20 years after implementation. Using data from Africa, Lee shows that quotas boost the share of women ministers with legislative experience— an important finding suggesting that after gaining access to the legislature via quotas, women gain opportunities to serve in cabinet in the next stage of their career. These studies underscore scholarly recognition of the importance of understanding the broader implications of quotas. Yet, existing work focuses on specific

regions, types of quotas, or measures of quota design, leaving open how the broader adoption of legislated gender quotas—and the magnitude of change they produce in women’s legislative representation—shapes women’s access to the executive across political contexts.

Do Gender Quotas Explain Women’s Access to Cabinet?

Gender quotas offer substantial promise for increasing women in the executive branch. Quotas are one of the most common electoral reforms to emerge in the past half century (Hughes et al. 2019). And today, half of countries across the globe have adopted some form of legal gender quota. Gender quotas are designed explicitly to increase women’s presence in the legislative branch by requiring that women occupy a specific share of positions on the ballot or in office (Hughes 2011). With global quota diffusion, women’s presence in parliaments has increased more than threefold, moving from only 8.3 percent in 1990 to 27 percent in 2024. As Clayton (2021, p. 236), describes it: “many quota-adopting countries have made remarkable leaps in very short periods of time, doubling or even tripling women’s legislative representation in the first post-quota election.” Given the power of quotas to increase women’s presence in the legislature, it stands to reason that if the share of women in parliament predicts women’s access to cabinets, then gender quotas should also increase the share of women in cabinets.

Nonetheless, there are reasons to believe that quotas could undermine women’s access to cabinets. Women elected via quotas can be sidelined in politics rather than promoted to executive power (Bauer and Britton 2006). Some find that women in legislatures are marginalized and have fewer leadership opportunities than their male counterparts (O’Brien et al. 2015). Others suggest that the women who come to power via quotas may differ significantly from women who come to power absent quotas: for example, they might lack broad political networks (Franceschet and Piscopo 2014). If so, they may not be incorporated into the cabinet. As Lee and Park put it: “While adopting gender quotas in Indonesia,

South Korea, and Taiwan has helped to boost the electability of women candidates, it has not necessarily led to the increasing supply of female political elites equipped with the power resources and political connections needed for other political positions” (Lee and Park 2018, 166). Quotas can also cause backlash (Clayton 2015), leading people to scrutinize their qualifications (Murray 2014) and eschew their leadership—particularly among male elites “who attribute women’s underrepresentation to choices made by individual women, rather than structural discrimination” (Franceschet, Krook and Piscopo 2012, 12).

Previous research thus leaves room for doubt about whether quotas accelerate women’s access cabinet. And, studies examining the ‘spillover’ effects of quotas find mixed evidence regarding the extent to which quotas can have positive effects on women’s leadership positions beyond those to which they directly apply. Some report positive effects, wherein quotas increase women’s access to leadership posts or increase the share of ministers with prior legislative experience. For example, O’Brien and Rickne (2016) examined quota adoption in Swedish municipalities and found women were more likely to be appointed to leadership posts in municipalities where quotas had a larger effect. Focusing on Ecuador, Basabe-Serrano and Perez (2022) show the likelihood of electing a woman mayor almost doubles in the aftermath of quota adoption. In an impressive study of approximately 5,000 ministers across 14 African countries, Lee (2024) shows that quota adoption boosts the share of cabinet ministers who previously had a seat in the legislature. Similarly in a study of 28 EU member states (1979-2022) Aldrich and Daniel (2025) show quotas intended for the EU Parliament also accelerate women’s careers into national, legislative, and executive offices.

Still, other studies find no such evidence. For instance, Kerevel (2019) finds that quotas have not increased women’s advancement to executive office when analyzing the career trajectories of quota-elected women in Mexico. Using a most similar case design to compare Italian regions with and without quotas, Weeks and Masala (2023) find little evidence that quotas had a spillover effect resulting in more women in leadership in the regional councils.

Research exploiting variation in quotas in Italy (Spaziani 2022), Spain (Bagues and Campa 2021), and France (Lassébie 2020) finds no evidence that quotas resulted in the election of more women mayors. Similarly, a careful examination of subnational politics in Sweden finds evidence of discriminatory promotion patterns such that women move up the political hierarchy at lower rates than equally qualified men (Folke and Rickne 2016).

We contribute to this research by examining the link between quotas and women in cabinets across countries and over time. Specifically, we evaluate the following hypothesis:
H1: Gender quota laws are associated with an increase in the share of women in cabinets.

Quotas and Cabinets: A Supply-Side Explanation

Our first expectation examines whether quotas are associated with increases in women's representation in cabinets. In addition to providing a systematic analysis of the relationship between quotas and women in cabinets, a second contribution of this study is to make progress on understanding the mechanism through which quotas work. Our main argument is that quotas increase the share of women in cabinet by increasing the number of women in parliament – i.e., by increasing the *supply* of women ministrables. Indeed, one of the most long-standing explanations for women's exclusion from political power is a supply side argument (Norris 1993; Paxton 1997). Scholars contend that women are more likely to be represented in office when there is a healthy supply of qualified women who are willing to oversee cabinet portfolios. In many countries, ministers are drawn from current or former MPs. Once in office, MPs can gain legislative experience, establish networks, grow their public image, and develop policy expertise – all important considerations for ministerial appointees. Consequently, quotas can act as a “positive disruptor” (Aldrich and Daniel, 2025, p. 18) to increase the share of women in cabinet by growing the pool of eligible appointees, the share of women in parliament.

If quotas increase the supply of women ministrables, then the relationship between

quotas and women in cabinet should be strongest where cabinet members are drawn from the legislature and where quotas are most effective at boosting women's presence in parliament. Further, the share of women in parliament should mediate the relationship between quota adoption and cabinet appointments. By contrast, if countries that adopt quotas are simply more inclined to appoint women to cabinet—that is, if the relationship is endogenous – a similar association could appear across institutional contexts, even where quotas have limited effects on women's parliamentary representation, though its strength may vary with pro-female elite preferences. We next consider the main observable implications of our supply-side argument, including variation in regime type and quota shocks. Then, we address potential demand-side dynamics.

Regime Type

A defining feature of regime types is the relationship between parliament and the executive branch. In Westminster systems, cabinet ministers are “drawn from (and remain) members of the legislature” (Stockemer 2017, 441). This means that parliament is the primary supply pool for ministers. Outside of Westminster systems it is not required that prime ministers (PMs) select appointees from parliament. Nonetheless, doing so allows PMs to use appointments to develop broader coalitions of support within parliament. This is important since they rely on parliament to stay in power. Thus, with some exceptions, cabinet nominees in parliamentary systems are typically drawn from parliament.

The path from parliament to cabinet is rarer in presidential and authoritarian systems. In presidential systems, legislative experience is viewed as one of many credentials that can enhance an appointee's appeal (Lee 2024; Nyrup et al. 2025). Instead, ministers are drawn from a wide range of experts, bureaucrats, and politicians (Lee and McClean 2022). Legislators appointed to cabinet portfolios are typically required to resign their seats.

In authoritarian systems the cabinet is drawn from the executive's inner circle of trusted

advisors. Although this may include select legislators, the legislature does not serve as a primary supply pool of potential nominees. Thus, if quotas lead to more women in cabinet by increasing the supply of eligible appointees, we should observe a stronger relationship in parliamentary systems, than in presidential systems or authoritarian regimes. *H2: The positive relationship between quotas and women in cabinet will be larger in parliamentary systems.*

Quota Shocks

Quotas vary considerably in the magnitude of their impact on women's numeric representation in parliament (Schwindt-Bayer 2009). Some quotas yield large gains, whereas others only moderately increase women's presence. We refer to this variation as differences in the magnitude of the quota shock—that is, the extent to which a quota alters the gender composition of parliament. If quotas influence cabinet composition by expanding the supply of women for cabinet, then we expect to observe a strong link between the size of the quota shock and women's presence in cabinets. The more quotas increase the share of women in office, the larger the pool of eligible women ministers. Consistent with this logic, for example, O'Brien and Rickne's (2016) analysis of the Swedish Social Democratic Party finds that the quotas had an 'acceleration effect' on the selection of women in local leadership roles, especially when they effectively expanded the pool of female candidates. But, when quotas are less successful at transforming the composition of parliament, and result in only small increases, this in turn limits the supply of women. Thus, to the extent that quotas increase women's presence in cabinet via supply side factors, we anticipate the relationship will be larger in countries with bigger quota shocks. *H3: The positive relationship between quotas and women in cabinet will be stronger where quota shocks are larger.*

Mediating Mechanism

The preceding hypotheses identify the context in which quotas are most likely to increase women's cabinet appointments. They do not, however, directly test how quotas exert that influence. If quotas expand women's cabinet representation by increasing their numbers in parliament, as the supply-side logic suggests, then we should also observe that the share of women in parliament mediates the relationship between quotas and cabinet appointments.

H4: The relationship between quotas and women in cabinets will be explained (mediated) by the share of women in parliament.

Isolating Potential Demand-Side Factors

Gender quotas might also work by increasing citizens' and elites' demand for women in cabinet. Indeed, there is evidence that chief executives respond to such demands when forming cabinets (Goddard 2019; Escobar-Lemmon and Taylor-Robinson 2016), and they use cabinet appointments to adapt to "changes in the political environment throughout their term" (Martinez-Gallardo 2014, p. 6). For citizens, debates surrounding quota design and implementation heighten public awareness of women's underrepresentation, and quota adoption can foster "societal attitudes in favour of higher numbers of female politicians" (Fernández and Valiente 2021, p.367). Even if citizens are unaware of quotas, simply seeing more women in politics can alter their "mental template" of what a leader looks like (Taylor-Robinson and Geva, 2023), reduce gender bias (Beaman et al. 2009), and foster more gender egalitarian attitudes (Alexander 2012). These shifts in public opinion may constitute an important source of demand for women's leadership.

Quotas can also heighten elite demand for women in cabinet. Quotas disrupt the recruitment of homogenous candidates from insular political networks, working to redefine elites' "perceptions of an ideal candidate" (Barnes and Holman 2020, 1271). They institutionalize norms and practices that promote greater inclusion (O'Brien et al., 2015). And, women elected via quotas may work together, "forming strategic coalitions" and

lobbying party leaders to incorporate more women into the cabinet (Krook and O'Brien 2012, 948).

Yet, quotas can also provoke resistance. In authoritarian regimes, quotas may serve strategic or international signals (Bush, Donno and Zetterberg 2024; Bush and Zetterberg 2021; Donno, Fox and Kaasik 2022). Quotas may trigger backlash among elites or citizens who perceive them as threatening or unfair (Clayton 2018, 2015) especially where quotas are foreign-imposed, associated with an unpopular regimes, or weakly enforced (Bush 2011). Consistent with this, Kim and Fallon (2023) show quotas raise public approval of women politicians in democracies but not authoritarian regimes, underscoring that demand effects vary by regime type.

While there are good theoretical reasons to expect demand-side effects, disentangling supply- and demand-side effects presents an empirical challenge. Quotas increase the pool of women ministrables, who may themselves generate demand for women in senior cabinet roles (Krook & O'Brien 2012). We cannot rule out these demand effects operating simultaneously with the supply mechanism we theorize. As a result, some of what appears to be demand may instead reflect second-order supply effects. Additionally, it is difficult to rule out the possibility that the same underlying pressures driving quota adoption also influence women's appointments—that is, the relationship is endogenous. Despite these empirical challenges, here, we offer one way to test for potential demand-side effects by considering the regime context.

As previously discussed, in parliamentary systems, supply-side effects are more likely: women in legislatures are often simultaneously eligible for cabinet positions. In contrast, in presidential systems, legislators must typically resign their seats to join the cabinet, so quota-induced increases in parliament do not immediately expand cabinet supply. If quota implementation nonetheless leads to an increase in women in cabinets in presidential systems, this is consistent with evidence that domestic demand—from citizens and political elites—is

driving the change. This is especially true in the context of low-shock quotas.

Presidential regimes also offer a cleaner test of demand-side effects compared to authoritarian regimes. Because in authoritarian regimes quotas are typically adopted by the same autocrats who control years of subsequent cabinet appointments, there may be unobserved confounders that simultaneously drive the adoption of quotas and the appointment of women. In presidential regimes, by contrast, quotas are often initiated in the legislature whereas cabinet appointments are made by the president (Gatto 2025). This separation of powers, in addition to electoral turnover of political leadership in presidential systems, makes it less likely that quota adoption and cabinet appointments are subject to the same endogeneity concerns that arise in authoritarian regimes.

In short, presidential systems present the clearest opportunity to observe quota-induced demand effects, both because supply-side channels are weaker and because democratic executive turnover opens windows for responsive appointments. Thus, if quotas increase women in cabinet by increasing domestic (elite or citizen) demand, we expect to observe a positive relationship between quotas and women in cabinet in presidential regimes. *H5: We expect to observe a positive relationship between quotas and women in cabinet in presidential regimes.*

Empirical Strategy

We employ multiple empirical approaches to examine the relationship between quotas and women's presence in cabinets. Each method has limitations. Rather than relying too heavily on any single approach, we adopt a multi-method strategy that allows us to cross-validate our findings, with evidence consistently pointing in the same direction across methods.

In the next section we describe our data and variables. Then, our analysis proceeds in three stages. First, we present descriptive figures of the data, before and after quota adoption, for visual inspection of trends. Second we use an event-study specification, following Callaway and Sant'Anna (2021), which we compare to a traditional two-way fixed effects model, to test the parallel trends assumption. To test H1-H3 and H5, we present

both traditional two-way fixed effects (TWFE) estimates and the robust difference-in-differences (DiD) estimator developed by Callaway and Sant’Anna (2021). The TWFE model includes country and year fixed effects, controlling for any country-specific omitted variables that are constant over time, such as historical gender norms, institutional context, and religious context, as well as any group-invariant trends over time, such as global economic conditions. It provides a conventional baseline widely used in the literature; however, given the negative weights issue identified in two-way fixed effects estimation (Imai and Kim, 2021), we also provide models using the robust estimator proposed by Callaway and Sant’Anna (2021). The CS approach is well-suited to settings with multiple time periods and staggered treatment adoption, and allows for conditioning on covariates. At the third stage, we conduct causal mediation analysis to formally assess whether women’s parliamentary representation mediates the effect of gender quota laws on women in cabinet (H4).

Data and Variables

We build a comprehensive, global dataset on the share of women in cabinets and gender quota laws by merging cabinet data from WhoGov (2020) with quota data from QAROT (2019), which we extended and updated from 2015 to 2021.³ We restrict the data to begin in 1990, before which gender quota laws are very rare. The final sample consists of 5,246 country-year observations: 168 countries from 1990 to 2021. We use annual data because cabinet composition can change at any time, not just during elections; leaders can appoint or dismiss ministers strategically throughout their terms (Annesley et al. 2019, 19). While annual data may capture smaller incremental changes than election-cycle aggregates, we verify findings using election-cycle–based measures (Appendix Tables D9-D10).

The main dependent variable measures the share of women appointed to the cabinet.

³Updated using information obtained from the Gender Quota Database (International IDEA, 2023).

Following Nyrup, Yamagishi and Bramwell (2023), our measure includes all full-ranking cabinet members. Our key independent variable is quota law, coded as a binary variable equaling 1 for each country-year after implementation of a quota law, and 0 otherwise.

To examine effects by regime type, we use data from Bjørnskov and Rode's (2020) updated version of Cheibub, Gandhi and Vreeland's Democracy-Dictatorship dataset (2010). Authoritarian regimes include autocracies with no party, one party, or multiple parties in government; parliamentary democracies are democracies where the government is responsible to the legislative assembly (i.e., parliamentary and mixed or semi-presidential systems), while presidential democracies are not. The categories are mutually exclusive.

To strengthen the causal interpretation of our results, we carefully account for potential confounding variables—factors that could influence both quota adoption and women in cabinets. First, thinking about political institutions, women's political inclusion and quota adoption are often justified by democratic principles (Franceschet and Piscopo, 2013). Following Nyrup, Yamagishi and Bramwell (2023), we use a stock variable (0 least democratic to 1 the most democratic history) to measure history of democracy.⁴ Next, with respect to socio-economic context, women's labor force participation can expand the supply of women ministrables and generate societal pressure for quota adoption (Inglehart and Norris, 2003). We use female labor force participation from the World Bank to account for these dynamics. Turning to selectors, owing to their more inclusive political ideologies, left-leaning selectors are often associated with the appointment of more women to cabinet (O'Brien et al., 2015; Goddard, 2021), and they may also be more inclined to oversee the adoption of a gender quota. Leftist parties are significantly more likely to implement internal party quotas than those on the right (Caul, 2001), which can generate “quota contagion” at the national level (Meier, 2004). We use the Database of Political Institutions to identify whether the chief

⁴Democracy scores from the V-Dem project's Polyarchy measure are summed from 1900 (or, if the country gained independence after the year 1900, the first year) to the given year, applying a 95% annual depreciation rate.

executive hails from a left-wing party—i.e., a communist, socialist, social democratic, or other leftist parties (Beck et al., 2001). Women selectors are likewise sometimes shown to tap more women for their cabinets (Reyes-Housholder, 2016); and, women leaders may be more likely to adopt gender quotas. We thus account for the presence of a women leader using the variable Female Leader derived from the WhoGov data (2020).⁵ See Table A1 for summary statistics of all variables included in the analysis. Table A2 reports the distribution of cases across system types (presidential, parliamentary, and authoritarian).

Results: Gender Quotas Increase Women in Cabinets

Descriptive Trends

We begin with a descriptive overview of quota implementation and women's cabinet representation. Figure 1 Panel A shows the share of women in cabinets in countries that implemented a quota, centered at year 0 (the year of implementation). We fit separate linear trends for pre- and post-quota periods. Pre-quota, the share of women in cabinets slowly increases over time. The mean rate of change (slope) over the twenty-year period before quotas are implemented is 0.26 percentage points per year. Post-quota, the share of women in cabinets jumps by nearly 2 percentage points, and the slope of the trend line rises to 0.36 percentage points, representing an increase of nearly 40 percent on the pre-quota trend.

We argue that quota laws will improve women's representation in cabinets especially when the quota works well to increase the share of women in parliaments. Figure 1 Panel B provides descriptive evidence in favor of this argument by dividing countries into those that see a large 'shock' to women in parliament on implementation of the law, versus those that do not. High shock countries are those with an above-average change in the first election post-quota, and low shock countries are those with a below-average change in the

⁵WhoGov relies on the Archigos dataset to code leadership, which identifies the effective leader as the person that de facto exercised power in a country (Goemans, Gleditsch and Chiozza 2009).

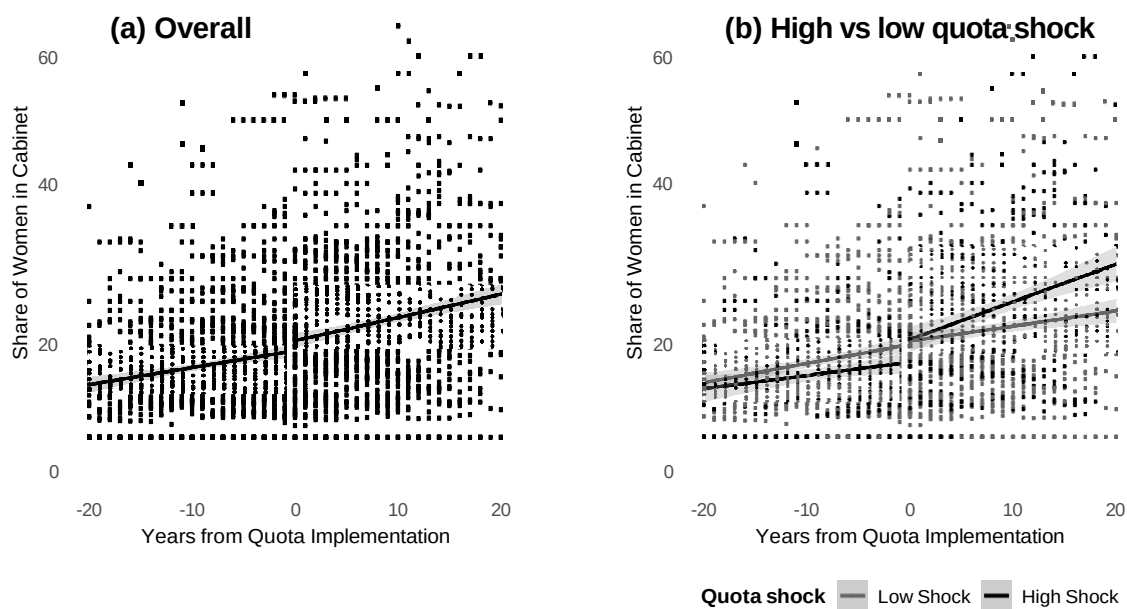


Figure 1: Mean share of women in cabinet before and after quota implementation
Alt text: The figure shows the mean share of women in cabinet before and after quota implementation overall (on the left) and by size of quota 'shock' or effectiveness at increasing the share of women in parliament (on the right).

first election post-quota.⁶ The figure shows that trends in the share of women in cabinets are very similar pre-quota law implementation for low and high shock countries. However, on implementation a large jump in the share of women in cabinets is observed for high shock, but not low shock, countries. The mean share of women in cabinets increases from 11.5 to 15.4 percentage points on quota implementation, and the rate of change (slope) is also higher (0.59%, compared to 0.21% pre-quota). Considering low shock countries, there is a smaller bump to the share of women in cabinets on quota implementation (increase of 0.8 percentage points), and the slope of the line remains similar across pre- and post-quota periods. Overall, this descriptive evidence is consistent with our expectations that quotas increase women in cabinet—particularly where they increase the supply of potential appointees.

Difference-in-Difference Estimators

To further examine the impact of quotas on women in cabinet, we employ a difference-in-differences (DiD) strategy. This approach compares the share of women in cabinet before and after a quota was implemented, relative to countries that did not implement such laws. The credibility of our estimates relies on the parallel trends assumption – that, absent quotas, ‘treated’ and ‘untreated’ countries would have followed similar trajectories in women’s cabinet representation. While quota adoption is not random, the legislative quotas we study target parliaments rather than cabinets and are generally adopted for reasons unrelated to executive appointments, reducing concerns about direct endogeneity. This assumption appears plausible, as shown in the DiD event study (Figure 2), which plots yearly treatment effects before and after quota implementation. The staggered timing of adoption across countries makes the event-study framework especially valuable, offering visual evidence of parallel pre-treatment trends and the dynamics of quota effects over time.

We present two versions of the event study. The main version uses the doubly robust

⁶The mean change to share women in parliament on quota implementation is an increase of 5.02 percentage points.

estimator proposed by Callaway and Sant’Anna (2021), which is well-suited for settings with multiple time periods and staggered treatment (quota) adoption, allowing for dynamic treatment effects and appropriate comparison groups. For comparison, we also present a second version using standard two-way fixed effects (TWFE). The x-axis of the plot is centered at the time of quota implementation (year 0), with effects shown up to 20 years before and after adoption. The y-axis displays the estimated share of women in cabinet positions. Results are also robust to using the Sun and Abraham (2021) estimator, which adjusts for variation in treatment timing by weighting cohorts accordingly (see Appendix Figure B2).

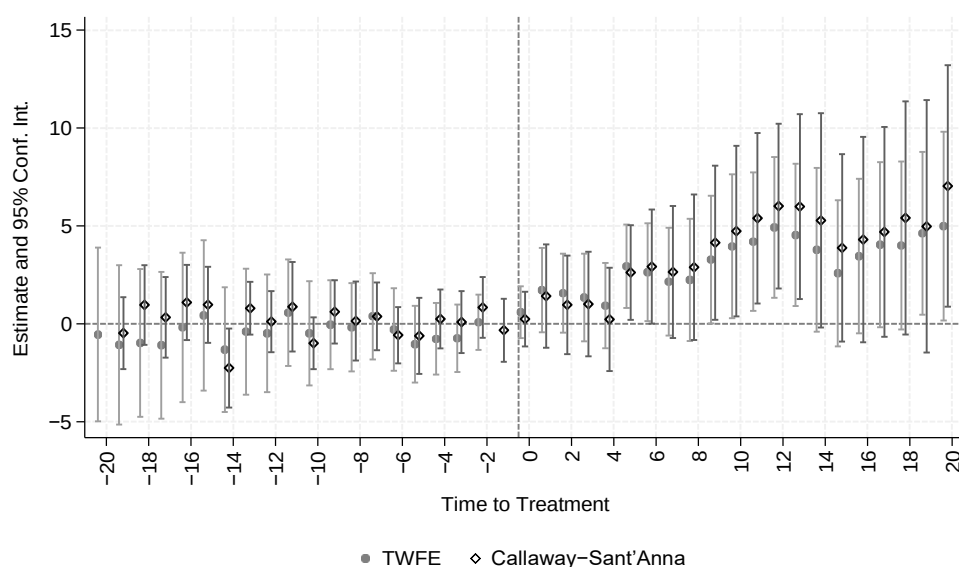


Figure 2: Event Study, Share of Women in Cabinets Centered on Quota Implementation
Note: Difference-in-differences event studies estimated using the `drdid` package in Stata. Both models (CS estimator and TWFE) include all controls and standard errors clustered by country.
Alt text: The figure presents an event study analysis of the share of women in cabinets centered on quota implementation, showing no evidence of pre-trends and an increase in the share of women in cabinets post-quota implementation.

Both event study models show that pre-treatment estimates are close to zero and statistically indistinguishable from it, supporting the parallel trends assumption. Post-treatment, we observe a positive and increasing trend in women’s cabinet representation, although confidence intervals widen over time—as expected—due to the declining number of countries with long post-treatment periods. Notably, the effect is more pronounced in electoral cycles following quota adoption, consistent with delayed institutional impact.

Visual evidence in Figure 2 supports the validity of our DiD design and suggests quota laws have a positive effect on increasing the share of women in cabinet. Joint pre-trend tests fail to reject the null of pretrends in most specifications (Appendix Table D1). However, as with all observational studies we are cautious in interpreting the results as causal.

Having offered support for the parallel trends assumption, Figure 3 presents estimates from both two-way fixed effects models and the average treatment effects on the treated (ATT) estimated using the robust DiD estimator by Callaway and Sant'Anna (2021). Following their recommendation, we use a doubly robust estimation method and a uniform confidence band. This method addresses well-known limitations of two-way fixed effects models in settings with staggered treatment adoption. It allows the parallel trends assumption to hold both unconditionally and conditionally on covariates. This is beneficial because, in situations where the treatment assignment might be related to observed characteristics, it improves identification by controlling for selection bias. Further, we add covariates to help reduce residual variance, leading to more precise estimates.

Our data follows a staggered adoption structure without proper treatment reversal, where countries implement quotas at different points in time and, with few exceptions, stay treated for the rest of the analysis.⁷ Countries that implemented quotas before 1990 are excluded from the analysis. Our control group is countries that have never or not yet implemented quotas. All models use clustered standard errors at the country level.

Figure 3 presents models with and without relevant covariates (see Online Appendix section C for full results). Here we show the estimates of the group-time average treatment effects, which estimate a separate average treatment effect for each group that is affected by the treatment (quota implementation) at different points in time. The estimates are

⁷The results are robust to excluding any country from the sample, including those rare cases which implement a quota and later repeal it – e.g., Italy (see Appendix Figure D1).

then aggregated to understand how the average treatment effects vary with the duration of exposure. This is also useful because it can be interpreted similarly to the TWFE estimator (Callaway and Sant’Anna, 2021, p. 208).

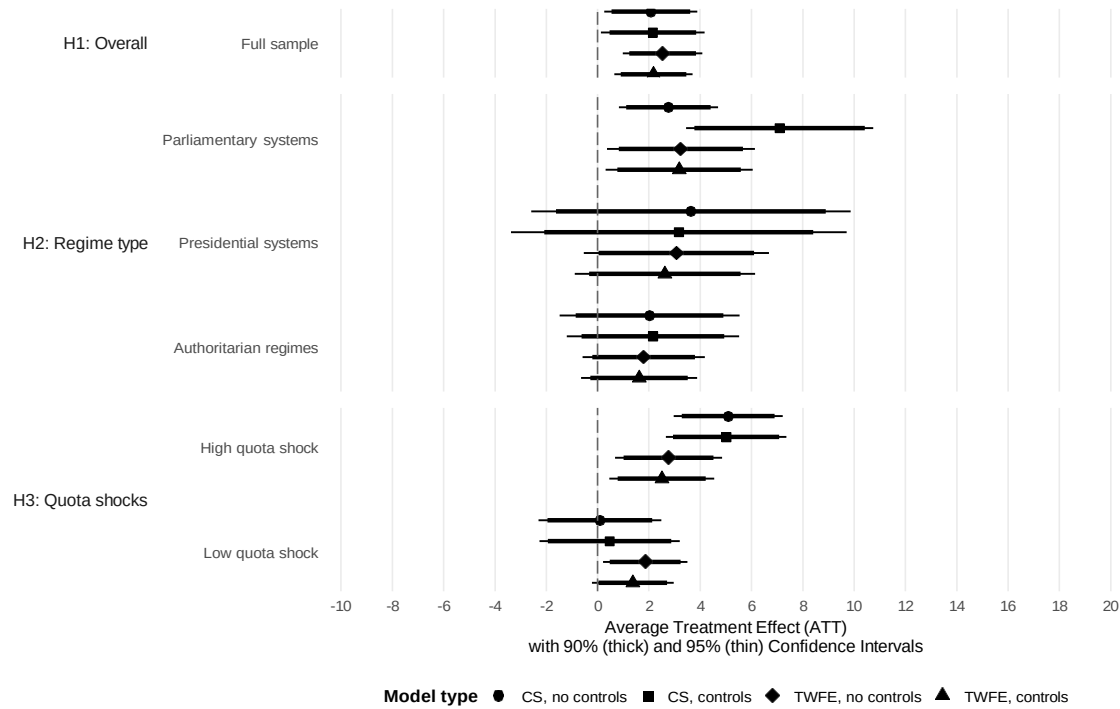


Figure 3: Effect of Quota Implementation on Share of Women in Cabinet
Note: Difference-in-differences and TWFE estimates of the average treatment effect on the treated, standard errors clustered by country. CS analysis was conducted using the did package in R (Callaway and Sant’Anna, 2021).
Alt text: The figure presents estimates of the effect of quota implementation on the share of women in cabinet from difference-in-differences and two-way fixed effects analyses, with and without controls.

The main results show that gender quota laws significantly increase women’s cabinet representation. Specifications using the full dataset show the estimated treatment effects range from 2.1 to 2.6 percentage points. This effect is meaningful relative to the sample mean of 14%, amounting to a 15 percent relative increase. These results provide strong support for H1: quota adoption leads to a rise in the share of women in cabinet positions.

We next examine whether the effects of quotas differ by regime type. If quota laws operate through expanding the pool of experienced female legislators, we expect stronger

effects in parliamentary systems, where ministers are often drawn from parliament. Consistent with H2, we find significant and positive effects in parliamentary democracies across specifications, while estimates for presidential systems and authoritarian regimes are positive but not statistically significant. We thus fail to find support for H5—the demand-side hypothesis. Instead, findings are consistent with a supply-side mechanism—where quotas increase the availability of qualified women—rather than a demand-side explanation alone, such as a shift in elite or public preferences for appointing women.

Finally, we examine variation by the magnitude of quota shocks in increasing women’s parliamentary representation. Following Clayton and Zetterberg (2018) and Latura and Weeks (2023), we distinguish between “high” and “low” shock quota adoptions, based on whether the initial increase in women’s parliamentary share exceeds the sample mean (5 percentage points). In line with H3, we find that cabinet gains are concentrated in countries with high shocks. Among high-shock countries, quotas increase the share of women in cabinets by between 2 and 5 percentage points on average, depending on the specification, while effects in low-shock countries are smaller and typically statistically indistinguishable from zero.⁸ We further investigated how the size of quota shocks to women in legislatures influences women in cabinet within regime types. This enables us to better isolate cases where we might observe independent demand-side effects (i.e., low-shock in presidential systems). High-shock quotas are associated with significant increases in women’s cabinet representation within both parliamentary and authoritarian systems, while low-shock quotas yield smaller effects, reaching significance only in parliamentary democracies (Appendix Tables D11-D12). These results continue to offer evidence for the supply-side explanation, while suggesting little support for the idea that quota-induced demand alone increases women’s access to power.

We conduct sensitivity checks to assess the robustness of our DiD estimates. Our main

⁸Low-shock quota implementation is positive and significant at the 0.05 level in the TWFE model without controls; however, this finding is not robust to the inclusion of controls.

models include only the most important confounders—those plausibly linked to both quota adoption and women’s cabinet representation—because adding too many controls can affect model performance, particularly in sub-sample analyses where data are sparser. First, we test whether results hold when including additional variables that may affect women’s appointments: party institutionalization, corruption, party-level quotas, and wartime status (Appendix Table C4). We further examine the sensitivity of our findings by re-estimating CS models while restricting the control group to not-yet-treated units (Appendix Table D3), and conducting leave-one-out analyses to ensure estimates are not driven by single-country outliers (Appendix Figure D1). Our results remain consistent across these sensitivity checks.

Mediation Analysis

The descriptive trends and DiD analysis support our expectation that quota laws increase the share of women in cabinets. Further, we find that this relationship is strongest in contexts where supply-side dynamics are most likely to drive cabinet appointments. Our supply-side argument implies a specific causal pathway: that quotas expand the pool of eligible women by boosting their representation in parliament, which in turn increases their likelihood of cabinet appointment. Thus, the final stage of our analysis employs a causal mediation framework (Imai et al., 2011) to formally evaluate whether the effect of quota laws on women’s cabinet representation is mediated by the share of women in parliament.

While we are cautious about making strong causal claims, this approach helps us explore potential mechanisms consistent with our theoretical expectations. We use the mediation package in R, which produces average total effects, and further decomposes these into average causal mediation effects and average direct effects. We first fit the mediator model (regressing the share of women in parliament, our mediator, on the share of women in cabinets, including the full list of controls and country and year fixed effects) and the outcome model (regressing quota law and the share of women in legislators on the share of women in cabinets, including

controls and fixed effects). Then, the package uses the algorithm specified by Imai et al. (2010) to generate the mediation and direct effect estimates.

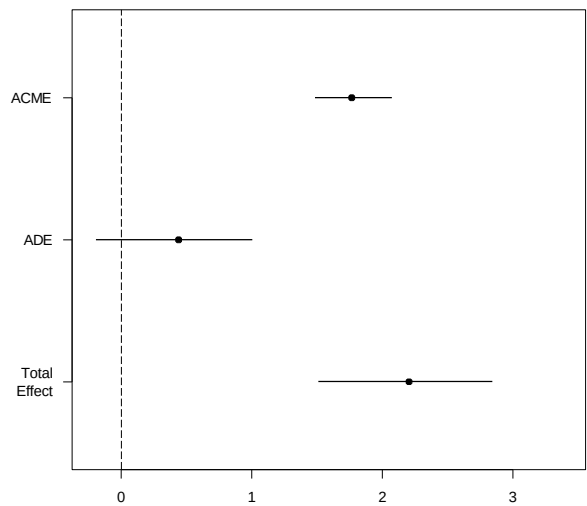


Figure 4: Mediation Analysis

Note: Causal mediation analysis was conducted using the mediation package in R (Tingley et al., 2014).
Alt text: The figure presents results of causal mediation analysis, showing that 82 percent of the total effect of quota laws on the share of women in cabinet is mediated by the share of women in parliament (evidence for the supply-side mechanism).

Figure 4 shows the results of the mediation analysis, with 95% confidence intervals. The total effect of a 2.2 percentage point increase in share of women in cabinets is plotted on the bottom row of Figure 4. The top two rows of the plot decompose the total effect into the average causal mediation effect (ACME, shown in the top row) and the average direct effect (ADE, shown in the middle row). The ACME is a 1.77 percentage point increase in the share of women in parliament, significant at conventional levels, while the ADE is 0.44, and not significant. The analysis reports that 82% of the total effect of quota laws on the share of women in cabinets is mediated by the share of women in parliaments. This offers strong evidence for the supply-side mechanism, in line with H4. Quotas increase the share of women in cabinets by boosting the share of women in the pipeline to power.

Where Are Women Appointed?

While our core analyses demonstrate that gender quotas increase women's overall cabinet representation, not all cabinet posts are created equal. A small set of high-prestige portfolios—such as finance, defense, foreign affairs, and interior—offer greater influence, visibility, and institutional power. This raises a critical follow-up question: *where* are women appointed? If gains are concentrated in low- and medium-prestige portfolios, this may reflect the persistent gendered norms that shape the distribution of executive power. Conversely, inclusion in high-prestige portfolios suggests that quotas can facilitate broader redistribution of political power. In this section, we examine the extent to which women's appointments after quota adoption reach into the upper echelons of cabinet.

Prior research offers competing expectations as to whether quota-induced gains may extend beyond low-prestige portfolios to more influential, inner-cabinet posts. On one hand, cabinet newcomers—regardless of gender—often begin in less prestigious roles (Lee and McClean, 2022). High-prestige appointments, by contrast, may require a track record in national politics—and this bar may be especially high for women. In their study of 27 European countries (1990–2018), Kroeber and Hüffelmann (2022) find that even highly qualified women in executive office wait significantly longer than men to reach top posts. Instead, women are more frequently appointed to portfolios aligned with traditional gender norms. If quotas expand the supply of eligible women without changing these norms or career pipelines, initial gains may be concentrated in lower-prestige, “entry-level” positions.

On the other hand, there are at least two reasons why quotas may expand women's access to high-prestige portfolios, both of which operate through the increased supply of women in parliaments. First, the share of women in parliament is the most consistent predictors of women's appointments to high-prestige portfolios (Krook and O'Brien, 2012; O'Brien et al., 2015; Escobar-Lemmon and Taylor-Robinson, 2016). Demand-side factors,

such as party ideology and electoral system, do not consistently predict access to these posts, suggesting supply is the driving force. Second, whereas men appointees are more likely to have strong party ties, women are more likely to be well-credentialed, holding relevant professional experience and educational expertise (Armstrong et al. 2024, Escobar-Lemmon and Taylor-Robinson 2016). Women elected under gender quotas also tend to be highly qualified (Aldrich and Daniel, 2024; Besley et al., 2017; O'Brien and Rickne, 2016). Thus, quota adoption may unlock a previously overlooked pool of exceptionally credentialed women, accelerating some into high-prestige cabinet roles.

Combined, these perspectives offer competing expectations. As such, we do not advance a specific hypothesis. Instead our exploratory analysis offers suggestive evidence about the depth of women's incorporation into executive office.

Analyzing Quota Effects by Portfolio Prestige

We employ our preferred CS estimator to examine the effect of quotas on the share of women in low-, medium-, and high-prestige posts (Figure 5 and Appendix Tables C2-C3). These variables measure the percentage of women appointed to portfolios that are categorized as low-prestige (e.g., youth, tourism, women's affairs, without portfolio), medium-prestige (e.g., agriculture, labour, justice) and high-prestige (e.g., defence, finance, foreign affairs). The prestige typology we use was developed by Krook and O'Brien (2012) and operationalized by Nyrup and Bramwell (2020) in the WhoGov data. For comparison, we also present results from traditional two-way fixed effects (TWFE) models in Appendix Tables C5-C9; while they differ somewhat in magnitude, especially for low-prestige portfolios, they generally point in the same direction for medium- and high-prestige roles.

Looking at the share of women, the left panel of Figure 5 shows that results are not confined to a single group of portfolios. Although estimates are not statistically significant for the share of women in low- or medium-prestige portfolios, the imprecision observed likely

stems from the fact that low- and (some) medium-prestige portfolios are not consistently present across all country-years. While virtually every country maintains high- and (some) medium-prestige portfolios, low-prestige portfolios are less common. Approximately 20 per cent of country-years do not appoint any ministers to low-prestige roles, which reduces the number of observations and contributes to greater uncertainty in the estimates.

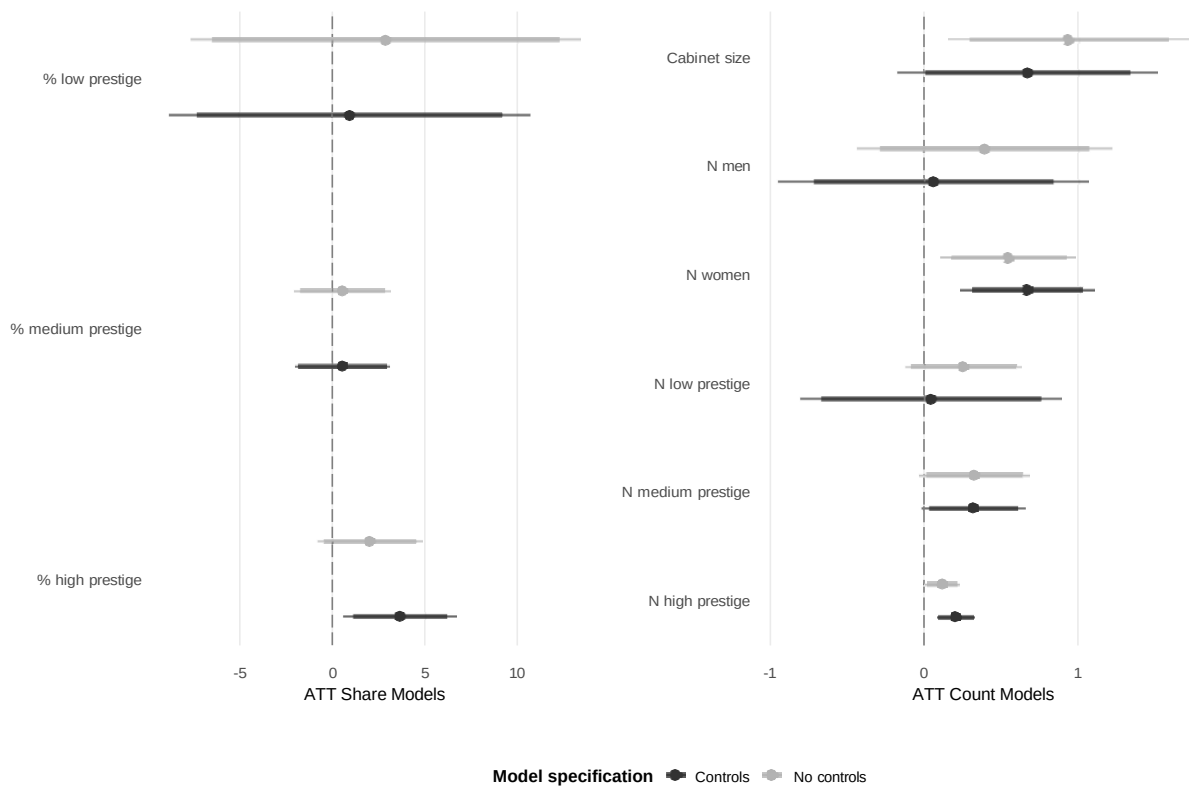


Figure 5: Effect of Quota Implementation by Prestige of Portfolio

Note: CS difference-in-differences estimates of the average treatment effect on the treated, standard errors clustered by country. Analysis was conducted using the did package in R (Callaway and Sant’Anna, 2021). Alt text: The figure shows the effect of quota implementation by prestige of portfolio (low, medium, and high), considering both the share (on the left) and number (on the right) of women in cabinets, as well as overall cabinet size and the number of men.

For high-prestige portfolios, quotas show a statistically significant positive effect only when controls are included (ATT = 3.7pp), suggesting quotas may expand women’s access to powerful posts. Analysis using our preferred CS approach finds that these gains are driven by women’s appointments to foreign affairs portfolios and are not consistently observed for other high-prestige portfolios (Appendix Table D5).

A potential concern is that governments respond to quotas by expanding cabinet size, allowing more women to enter without displacing existing members. While this would still reflect greater inclusion, it could dilute women's power if new positions are concentrated in lower-prestige portfolios. In such cases, share measures may understate the true extent of change: if new, low-prestige roles are added primarily for women, the proportion of women in cabinet may rise only modestly—even as their absolute numbers increase. Therefore, we also estimate DiD models predicting overall cabinet size, the number of women and men ministers, and the number of women by prestige (right panel of Figure 5). The results show that quotas are indeed associated with an increase in total cabinet size ($ATT = 0.8$, $p < 0.05$), although this falls below conventional significance levels on the inclusion of controls ($ATT = 0.7$, $p < 0.1$). And, in line with our main results, the number of women ministers increases significantly—by 0.5–0.6 on average ($p < 0.05$). Importantly, this increase does not come at men's expense: quotas are not associated with any decrease in the number of male ministers. In fact, the ATT estimates for men are positive (though insignificant), reinforcing the view that women's gains are driven by expansion rather than substitution.

Disaggregating these effects by portfolio prestige, the count models in Figure 5 offer more evidence of cabinet expansion. The largest point estimates appear in low- and medium-prestige ministries, and when we combine these two categories to address sparse data in low-prestige portfolios, the number of women appointed to low–medium posts increases significantly ($p < 0.05$), even though the corresponding share-based estimates remain small and statistically insignificant (see Appendix Tables D7–D8). This divergence – significant growth in counts without comparable growth in shares – is consistent with a cabinet-expansion mechanism in which governments add new, lower-prestige positions following quota adoption. Turning to high-prestige portfolios, the estimated increase in the count of women ministers is smaller in absolute terms ($ATT = 0.2$ with controls) but statistically significant at the 1 percent level. Thus, while some of the increase in women's representation reflects cabinet expansion – particularly in less prominent roles – quota laws

also facilitate meaningful inclusion in powerful ministries.

Conclusion

Most countries now apply some form of electoral gender quota as a policy lever to boost numbers of women in parliaments. Until now, we have yet to understand the potential follow-on impacts of these popular policies on the appointment of women to even higher positions of power – cabinet posts. To address this important question, we use a difference-in-differences approach in a global sample of countries from 1990 to 2021. Our study finds that, on average, quota laws are associated with a 2.1-2.6 percentage point gain in the share of women in cabinets. The results offer further evidence that quotas can have positive spillover effects on women’s promotion to leadership roles (O’Brien and Rickne 2016; Lee 2024).

Our study shows that quotas work by boosting the supply of women in the ministerial candidate pool, and this translates into more women in cabinet especially in parliamentary systems, where cabinets are often selected from the parliament, and in countries where the quota leads to a large shock of women in parliament. We find little evidence to support the demand-side mechanism *alone* – that quotas work by raising the salience of gender equality in the public eye and to party leaders, even when women’s supply is not boosted and parliament is not a pipeline to cabinet (i.e., low shock quotas in presidential and authoritarian systems).

Our disaggregated analysis finds that quota-driven gains are not confined to the margins of government. Women’s numbers rise most in low- and mid-prestige portfolios, but we also detect a statistically significant, albeit smaller, increase in top-tier posts. Cabinets therefore integrate women at multiple ranks rather than relegating them exclusively to peripheral portfolios. Crucially, the influx of women does not crowd men out: the total number of male ministers remains stable, indicating that many governments respond to quotas by creating ministries instead of replacing (mostly male) incumbents. This pattern is consistent with the

idea that elites accommodate gender reforms in ways that limit disruption to existing power hierarchies, a tendency noted in earlier work on legislative seat expansion and the phased implementation of quotas (Weeks 2018; Zetterberg et al. 2022).

Our study has several limitations. First, the cross-national DiD approach we use enables broad generalization, but necessarily trades off internal validity for scope. Given the difficulty of isolating demand-side effects from supply-side changes, and the potential for unobserved institutional or cultural factors to confound estimates, future research could use alternative designs, such as within-country longitudinal studies (pre- and post-quota adoption) or qualitative approaches (surveys of public opinion or interviews with party leaders) to examine how quotas might shift attitudes over time. Quasi-experimental approaches could also be informative; for instance, by examining whether quotas increase women in cabinets even among parties that opposed quota adoption but were nonetheless subject to the law, which could help disentangle supply- and demand-side effects.

Second, our analysis of subgroups, including by regime type and magnitude of quota shock, is inherently less precisely estimated than the larger full-sample models. As post-quota country-year observations grow, future research will be able to better assess how the effects of quotas are conditioned by political context and magnitude. Future research could also examine how quotas operate under different contexts. For example, labor market gender gaps and women's access to elite professional networks may influence whether increases in women legislators translate into a larger pool of qualified ministerial candidates—and which portfolios they are likely to occupy. Legislative rules and norms may also shape quota spillovers into executive office. In contexts with strong seniority norms or high reelection rates, quotas may take longer to increase women's cabinet representation, whereas in legislatures with weaker seniority norms or higher turnover, quota adoption may accelerate women's access to executive positions. Investigating these mechanisms could help clarify when and where quotas are most effective.

The policy significance of these appointments should not be underestimated. Cabinets function as collective decision-making bodies, shaping national policy and representing the face of the government. As such, *any* increase in women's presence in cabinet represents a meaningful step toward gender equality in governance. Even ministries considered less prestigious—such as portfolios for women, family, or youth—shape policy direction on gender equality, work–family reconciliation, and social protection. A fruitful direction for future research is to investigate how gender quotas and their associated boosts to women in cabinet influence the substantive representation of women's interests, from child care and parental leave policy (see Atchison and Down 2009 and Weeks 2022 for studies of OECD countries) to violence against women, the gender pay gap, reproductive rights, and other equality issues. Further research should investigate the extent to which seeing women in visible cabinet posts can impact the symbolic representation of women, increasing for example women's political engagement and ambition (Alexander and Jalalzai 2020; Taylor-Robinson and Geva 2023). Finally, taking a cue from research on workplace diversity (e.g., Comi et al. 2020), there is much room for further study of how the added diversity that gender quotas bring to cabinets affects government performance and stability. The implementation of quotas may set into motion a range of cascade effects with important implications for gender equality.

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References

- Aldrich, Andrea S and William T Daniel. 2024. “Gender quota adoption and the qualifications of parliamentarians.” *The Journal of Politics* 86(2):798–803.
- Aldrich, Andrea S and William T Daniel. 2025. *Gender Quotas as Game Changers for the Recruitment, Selection, and Performance of Elected Politicians*. Oxford University Press.
- Alexander, Amy C. 2012. “Change in women’s descriptive representation and the belief in women’s ability to govern: A virtuous cycle.” *Politics & Gender* 8(4):437–464.
- Alexander, Amy C and Farida Jalalzai. 2020. “Symbolic empowerment and female heads of states and government.” *Politics, Groups, and Identities* 8(1):24–43.
- Annesley, Claire and Francesca Gains. 2010. “The core executive: Gender, power and change.” *Political Studies* 58(5):909–929.
- Annesley, Claire, Karen Beckwith and Susan Franceschet. 2019. *Cabinets, ministers, and gender*. Oxford University Press.
- Annesley, Claire and Susan Franceschet. 2015. “Gender and the executive branch.” *Politics & Gender* 11(4):613–617.
- Arriola, Leonardo R and Martha C Johnson. 2014. “Ethnic politics and women’s empowerment in Africa.” *American Journal of Political Science* 58(2):495–510.
- Atchison, Amy and Ian Down. 2009. “Women cabinet ministers and female-friendly social policy.” *Poverty & Public Policy* 1(2):1–23.
- Atchison, Amy L and Ian Down. 2019. “The effects of women officeholders on environmental policy.” *Review of Policy Research* 36(6):805–834.
- Bagues, Manuel and Pamela Campa. 2021. “Can gender quotas in candidate lists empower women?” *Journal of Public Economics* 194:104315.
- Barnes, Tiffany D and Mirya R Holman. 2020. “Gender quotas, women’s representation,

- and legislative diversity.” *The Journal of Politics* 82(4):1271–1286.
- Basabe-Serrano, Santiago and M Fabricio Perez. 2022. “Spillover effects of quota or parity laws: The case of Ecuador women mayors.” *Latin American Policy* 13(1):82–103.
- Bauer, Gretchen and Akosua Darkwah. 2022. “The president’s prerogative?” *Politics & Gender* 18(2):546–573.
- Bauer, Gretchen and Hanna E Britton. 2006. *Women in African parliaments*. Lynne Rienner.
- Bauer, Gretchen and Manon Tremblay. 2011. *Women in Executive Power*. Routledge London.
- Beaman, Lori, Raghavendra Chattopadhyay, Esther Duflo, Rohini Pande and Petia Topalova. 2009. “Does exposure reduce bias?” *The Quarterly journal of economics* 124(4):1497–1540.
- Beck, Thorsten, George Clarke, Alberto Groff, Philip Keefer and Patrick Walsh. 2001. “New tools in comparative political economy.” *the world bank economic review* 15(1):165–176.
- Bego, Ingrid. 2014. “Accessing power in new democracies: The appointment of female ministers in postcommunist Europe.” *Political Research Quarterly* 67(2):347–360.
- Besley, Timothy, Olle Folke, Torsten Persson and Johanna Rickne. 2017. “Gender quotas and the crisis of the mediocre man.” *American economic review* 107(8):2204–2242.
- Bjørnskov, Christian and Martin Rode. 2020. “Regime types and regime change.” *The Review of International Organizations* 15:531–551.
- Bush, Sarah Sunn. 2011. “International politics and the spread of quotas for women in legislatures.” *International Organization* 65(1):103–137.
- Bush, Sarah Sunn, Daniela Donno and P ä r Zetterberg. 2024. “International rewards for gender equality reforms in autocracies.” *American Political Science Review* 118(3):1189–1203.
- Bush, Sarah Sunn and P ä r Zetterberg. 2021. “Gender quotas and international reputation.” *American journal of political science* 65(2):326–341.

- Callaway, Brantly and Pedro HC Sant'Anna. 2021. "Difference-in-differences with multiple time periods." *Journal of econometrics* 225(2):200–230.
- Caul, Miki. 2001. "Political parties and the adoption of candidate gender quotas: A cross-national analysis." *The Journal of Politics* 63(4):1214–1229.
- Cheibub, Jos é Antonio, Jennifer Gandhi and James Raymond Vreeland. 2010. "Democracy and dictatorship revisited." *Public choice* 143:67–101.
- Claveria, Silvia. 2014. "Still a 'male business'? Explaining women's presence in executive office." *West European Politics* 37(5):1156–1176.
- Clayton, Amanda. 2015. "Women's political engagement under quota-mandated female representation." *Comparative Political Studies* 48(3):333–369.
- Clayton, Amanda. 2018. "Do gender quotas really reduce bias? Evidence from a policy experiment in Southern Africa." *Journal of Experimental Political Science* 5(3):182–194.
- Clayton, Amanda. 2021. "How do electoral gender quotas affect policy?" *Annual Review of Political Science* 24:235–252.
- Clayton, Amanda and P ä r Zetterberg. 2018. "Quota shocks: Electoral gender quotas and government spending priorities worldwide." *The Journal of Politics* 80(3):916–932.
- Comi, Simona, Mara Grasseni, Federica Origo and Laura Pagani. 2020. "Where women make a difference." *ILR Review* 73(3):768–793.
- Donno, Daniela, Sara Fox and Joshua Kaasik. 2022. "International incentives for women's rights in dictatorships." *Comparative Political Studies* 55(3):451–492.
- Escobar-Lemmon, Maria C and Michelle M Taylor-Robinson. 2016. *Women in presidential cabinets: Power players or abundant tokens?* Oxford University Press.
- Fernández, Juan J and Celia Valiente. 2021. "Gender quotas and public demand for increasing women's representation in politics: An analysis of 28 European countries." *European Political Science Review* 13(3):351–370.
- Field, Bonnie N. 2021. "Ministers, gender and political appointments." *Government and Opposition* 56(4):722–743.

- Folke, Olle and Johanna Rickne. 2016. “The glass ceiling in politics: Formalization and empirical tests.” *Comparative Political Studies* 49(5):567–599.
- Franceschet, Susan and Jennifer M Piscopo. 2013. “Equality, democracy, and the broadening and deepening of gender quotas.” *Politics & Gender* 9(3):310–316.
- Franceschet, Susan and Jennifer Piscopo. 2014. “Sustaining gendered practices? Power, parties, and elite political networks in Argentina.” *Comparative Political Studies* 47(1):85–110.
- Franceschet, Susan, Mona Krook and Jennifer Piscopo. 2012. *The impact of gender quotas*. Oxford University Press.
- Gatto, Malu. 2025. *Resistance to Gender Quotas in Latin America*. Oxford University Press.
- Goddard, Dee. 2019. “Entering the men’s domain? Gender and portfolio allocation in European governments.” *European Journal of Political Research* 58(2):631–655.
- Goddard, Dee. 2021. “Examining the appointment of women to ministerial positions across Europe: 1970–2015.” *Party Politics* 27(4):631–643.
- Goemans, Henk E, Kristian Skrede Gleditsch and Giacomo Chiozza. 2009. “Introducing Archigos: A dataset of political leaders.” *Journal of Peace research* 46(2):269–283.
- Homola, Jonathan. 2022. “The effects of women’s descriptive representation on government behavior.” *Legislative Studies Quarterly* 47(2):295–308.
- Hughes, Melanie M. 2011. “Intersectionality, quotas, and minority women’s political representation worldwide.” *American Political Science Review* 105(3):604–620.
- Hughes, Melanie M, Pamela Paxton, Amanda B Clayton and Pär Zetterberg. 2019. “Global gender quota adoption, implementation, and reform.” *Comparative Politics* 51(2):219–238.
- Imai, Kosuke and In Song Kim. 2021. “On the use of two-way fixed effects regression models for causal inference with panel data.” *Political Analysis* 29(3):405–415.
- Imai, Kosuke, Luke Keele and Dustin Tingley. 2010. “A general approach to causal mediation analysis.” *Psychological methods* 15(4):309.
- Imai, Kosuke, Luke Keele, Dustin Tingley and Teppei Yamamoto. 2011. “Unpacking the

- black box of causality: Learning about causal mechanisms from experimental and observational studies.” *American Political Science Review* 105(4):765–789.
- Inglehart, Ronald and Pippa Norris. 2003. *Rising tide: Gender equality and cultural change around the world*. Cambridge University Press.
- International IDEA, Stockholm. 2023. “Gender Quotas Database.”
- Kerevel, Yann. 2019. “Empowering women? Gender quotas and women’s political careers.” *The Journal of Politics* 81(4):1167–1180.
- Kim, Jessica and Kathleen M Fallon. 2023. “Making women visible: How gender quotas shape global attitudes toward women in politics.” *Politics & Gender* 19(4):981–1006.
- Kroeber, Corinna and Joanna Hüffelmann. 2022. “It’s a long way to the top: Women’s ministerial career paths.” *Politics & Gender* 18(3):741–767.
- Kroeger, Alex and Alice J Kang. 2022. “The appointment of women to authoritarian cabinets in Africa.” *Government and Opposition* pp. 1–24.
- Krook, Mona Lena and Diana Z O’Brien. 2012. “All the president’s men? The appointment of female cabinet ministers worldwide.” *The Journal of Politics* 74(3):840–855.
- Lassébie, Julie. 2020. “Gender quotas and the selection of local politicians: Evidence from French municipal elections.” *European Journal of Political Economy* 62:101842.
- Latura, Audrey and Ana Catalano Weeks. 2023. “Corporate board quotas and gender equality policies in the workplace.” *American Journal of Political Science* 67(3):606–622.
- Lee, Don S and Charles T McClean. 2022. “Breaking the cabinet’s glass ceiling.” *Comparative Political Studies* 55(6):992–1020.
- Lee, Don S and Soonae Park. 2018. “Democratization and women’s representation in presidential cabinets.” *Asian Journal of Political Science* 26(2):161–180.
- Lee, Dongil. 2024. *Dynamic Impact of Legislative Gender Quotas on Female Representation in Cabinets*. Technical Report WP24-12 KDI School of Public Policy and Management.
- Martínez-Gallardo, Cecilia. 2014. “Designing cabinets: presidential politics and ministerial instability.” *Journal of Politics in Latin America* 6(2):3–38.

- Meier, Petra. 2004. "The mutual contagion effect of legal and party quotas: A Belgian perspective." *Party Politics* 10(5):583–600.
- Murray, Rainbow. 2013. "Quotas, citizens, and norms of representation." *Politics & Gender* 9(3):304–309.
- Murray, Rainbow. 2014. "Quotas for men: Reframing gender quotas as a means of improving representation for all." *American Political Science Review* 108(3):520–532.
- Nyrup, Jacob, Carl Knutsen, Peter Langsæther and Ina Kristiansen. 2025. "Paths to power: A new dataset on social profile of governments." *British Journal of Political Science* 55:e129.
- Nyrup, Jacob, Hikaru Yamagishi and Stuart Bramwell. 2023. "Consolidating progress: the selection of female ministers in autocracies and democracies." *American Political Science Review* pp. 1–20.
- Nyrup, Jacob and Stuart Bramwell. 2020. "Who governs? A new global dataset on members of cabinets." *American Political Science Review* 114(4):1366–1374.
- O'Brien, Diana Z, Matthew Mendez, Jordan Carr Peterson and Jihyun Shin. 2015. "Letting down the ladder or shutting the door." *Politics & Gender* 11(4):689–717.
- O'Brien, Diana Z and Johanna Rickne. 2016. "Gender quotas and women's political leadership." *American Political Science Review* 110(1):112–126.
- Paxton, Pamela and Melanie M Hughes. 2015. "The increasing effectiveness of national gender quotas, 1990–2010." *Legislative studies quarterly* 40(3):331–362.
- Paxton, Pamela, Melanie M Hughes and Tiffany D Barnes. 2020. *Women, politics, and power: A global perspective*. Rowman & Littlefield Publishers.
- Reyes-Housholder, Catherine. 2016. "Presidentas Rise: Consequences for Women in Cabinets?" *Latin American Politics and Society* 58(3):3–25.
- Schwindt-Bayer, Leslie A. 2009. "Making quotas work: The effect of gender quota laws on the election of women." *Legislative studies quarterly* 34(1):5–28.
- Spaziani, Sara. 2022. "Can gender quotas break the glass ceiling? Evidence from Italian

- municipal elections.” *European Journal of Political Economy* 75:102171.
- Stockemer, Daniel. 2017. “The proportion of women in legislatures and cabinets: What is the empirical link?” *Polity* 49(3):434–460.
- Stockemer, Daniel and Aksel Sundström. 2018. “Women in cabinets: The role of party ideology and government turnover.” *Party Politics* 24(6):663–673.
- Sun, Liyang and Sarah Abraham. 2021. “Estimating dynamic treatment effects in event studies with heterogeneous treatment effects.” *Journal of econometrics* 225(2):175–199.
- Taylor-Robinson, Michelle M and Nehemia Geva. 2023. *The Image of Gender and Political Leadership: A Multinational View of Women and Leadership*. Oxford University Press.
- Tingley, Dustin, Teppei Yamamoto, Kentaro Hirose, Luke Keele and Kosuke Imai. 2014. “Mediation: R package for causal mediation analysis.” *Journal of statistical software* 59:1–38.
- Tripp, Aili Mari and Alice Kang. 2008. “The global impact of quotas: On the fast track to increased female legislative representation.” *Comparative Political Studies* 41(3):338–361.
- Weeks, Ana Catalano. 2018. “Why are gender quota laws adopted by men? The role of inter-and intraparty competition.” *Comparative Political Studies* 51(14):1935–1973.
- Weeks, Ana Catalano. 2022. *Making Gender Salient: From Gender Quota Laws to Policy*. Cambridge University Press.
- Weeks, Ana Catalano and Francesco Masala. 2023. “Gender Quotas and Women’s Inclusion in Local Politics in Italy.” *Legislative Studies Quarterly* 48(3):503–534.
- Zetterberg, Pär, Elin Bjarnegård, Melanie Hughes and Pamela Paxton. 2022. “Democracy and the adoption of electoral gender quotas worldwide.” *Comparative Politics* 54(2):327–348.

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Supplemental Material for ‘Gender Quota Laws and Women in Cabinets’ by Barnes, Venturini and Weeks

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A Summary Statistics

Table A1: Summary Statistics

Statistic	N	Mean	St. Dev.	Min	Max
Share women in cabinet	5,246	14.306	12.468	0.000	64.706
Share women in cabinet, low prestige	4,129	29.049	35.354	0.000	100.000
Share women in cabinet, medium prestige	5,239	14.958	15.009	0.000	100.000
Share women in cabinet, high prestige	5,244	7.564	14.300	0.000	100.000
Quota law	5,259	0.238	0.426	0	1
Quota law, high shock	5,259	0.087	0.283	0	1
Quota law, low shock	5,259	0.151	0.358	0	1
No. ministers	5,259	20.319	7.393	0	67
No. male ministers	5,259	17.488	7.109	0	63
No. female ministers	5,259	2.831	2.600	0	18
No. female ministers, low prestige	5,246	0.584	0.850	0	6
No. female ministers, medium prestige	5,246	1.970	2.053	0	20
No. female ministers, high prestige	5,246	0.361	0.680	0	4
Parliamentary democracy	5,259	0.352	0.478	0	1
Presidential democracy	5,259	0.201	0.401	0	1
Authoritarian regime	5,259	0.447	0.497	0	1
Stock of polyarchy (95%) _(t-1)	5,203	0.452	0.269	0.019	1.000
Women's labor force _(t-1)	5,114	49.050	15.738	5.922	90.466
Woman leader _(t-1)	5,224	0.050	0.219	0	1
Left executive	5,259	0.263	0.440	0	1
Political corruption _(t-1)	5,187	0.514	0.301	0.002	0.966
Any war	5,227	0.047	0.212	0	1
Party institutionalization	5,078	0.604	0.257	0.008	1.000
Party quota	5,259	0.112	0.316	0	1
Year	5,259	2005.740	9.160	1990	2021

Note: Table A2 shows the mean and standard deviation for each variable by regime type.

Table A2: Summary Statistics by Regime Type

Variable	Authoritarian	Parliamentary	Presidential
Share women in cabinet	9.926 (9.773)	18.319 (14.089)	16.983 (11.628)
Share women in cabinet, low	25.558 (33.286)	30.972 (37.244)	34.107 (35.989)
Share women in cabinet, medium	9.668 (10.807)	19.976 (17.384)	17.853 (14.704)
Share women in cabinet, high	4.641 (11.675)	10.472 (16.061)	8.947 (15.056)
Quota law	0.211 (0.408)	0.189 (0.392)	0.383 (0.486)
Quota law, high shock	0.075 (0.264)	0.068 (0.251)	0.149 (0.357)
Quota law, low shock	0.136 (0.343)	0.122 (0.327)	0.233 (0.423)
No. ministers	22.676 (7.407)	18.171 (6.099)	18.844 (7.891)
No. male ministers	20.31 (6.866)	14.95 (6.057)	15.663 (7.077)
No. female ministers	2.366 (2.529)	3.221 (2.669)	3.181 (2.466)
No. female ministers, low	0.593 (0.874)	0.526 (0.776)	0.667 (0.909)
No. female ministers, medium	1.628 (2.052)	2.289 (2.086)	2.169 (1.875)
No. female ministers, high	0.252 (0.616)	0.506 (0.776)	0.348 (0.579)
Stock of polyarchy (95%) _(t-1)	0.247 (0.138)	0.663 (0.241)	0.531 (0.196)
Women's labor force _(t-1)	47.424 (19.979)	50.117 (11.011)	50.747 (11.252)
Woman leader _(t-1)	0.018 (0.132)	0.074 (0.262)	0.081 (0.273)
Left executive	0.261 (0.439)	0.268 (0.443)	0.258 (0.438)
Political corruption	0.667 (0.223)	0.304 (0.279)	0.547 (0.272)
Any war	0.072 (0.259)	0.019 (0.138)	0.04 (0.195)
Party institutionalization	0.452 (0.223)	0.772 (0.205)	0.623 (0.209)
Party quota	0.074 (0.262)	0.079 (0.27)	0.255 (0.436)
N	2350	1851	1058

B Event Study

Unconditional Parallel Trends

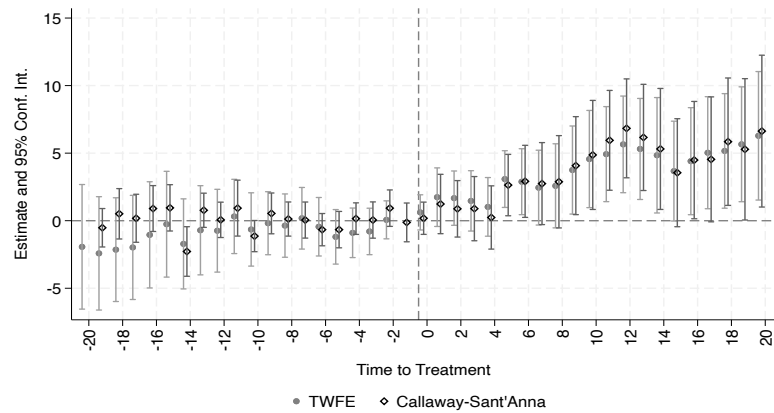


Figure B1: Event Study, Share of Women in Cabinets Centered on Quota Implementation
Note: Difference-in-differences event studies estimated using the `drdid` package in Stata. Both models (CS estimator and TWFE) include standard errors clustered by country.

Sun & Abraham (2020)

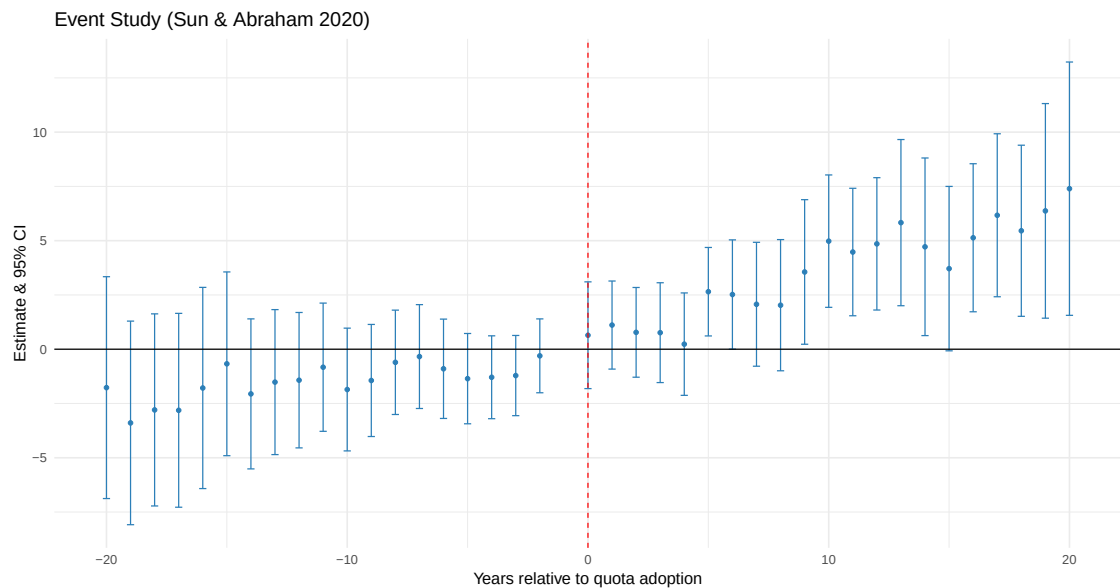


Figure B2: Event Study, Share of Women in Cabinets Centered on Quota Implementation
Note: Event study estimated using the `fixest` package in R. Standard errors clustered by country.

C Results Tables

Robust DID Models

Table C1: Effects of Quota Implementation on Share of Women in Cabinets by Model Type

Model	Estimate (SE)	Controls
<i>Overall (H1)</i>		
Full Sample	2.08** (0.93)	No Controls
Full Sample	2.16** (1.03)	Controls
<i>Regime type (H2, H5)</i>		
Parliamentary	2.77*** (0.99)	No Controls
Parliamentary	7.1*** (1.86)	Controls
Presidential	3.64 (3.18)	No Controls
Presidential	3.17 (3.34)	Controls
Authoritarian	2.03 (1.79)	No Controls
Authoritarian	2.16 (1.71)	Controls
<i>Quota shock (H3)</i>		
High Quota Shock	5.1*** (1.08)	No Controls
High Quota Shock	5.02*** (1.2)	Controls
Low Quota Shock	0.1 (1.22)	No Controls
Low Quota Shock	0.48 (1.4)	Controls

Note: Robust standard errors clustered around country in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Controls include (lagged) democratization (stock of polyarchy (95%)), women's labor force participation, left executive, and woman leader. For the subsample of authoritarian regimes, the estimates will not compute with full the covariates used in the rest of the models due to not enough control units for any group. For authoritarian regimes, therefore, controls include democratization and women's labor force participation only.

Table C2: Effect of Quota Implementation on Share Women in Cabinets, by Prestige of Portfolio

Model	Estimate (SE)	Controls
Low Prestige	2.88 (5.37)	No
Low Prestige	0.94 (4.97)	Yes
Medium Prestige	0.54 (1.33)	No
Medium Prestige	0.55 (1.3)	Yes
High Prestige	2 (1.44)	No
High Prestige	3.65** (1.55)	Yes

Note: Robust standard errors clustered around country in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C3: Effects of Quota Implementation on Size of Cabinets, by Gender, and by Prestige

Model	Estimate (SE)	Controls
Cabinet Size	0.94** (0.4)	No
Cabinet Size	0.67* (0.43)	Yes
Number of Men	0.39 (0.42)	No
Number of Men	0.06 (0.51)	Yes
Number of Women	0.54** (0.22)	No
Number of Women	0.66*** (0.22)	Yes
Low Prestige (Count)	0.25 (0.19)	No
Low Prestige (Count)	0.04 (0.43)	Yes
Medium Prestige (Count)	0.32* (0.18)	No
Medium Prestige (Count)	0.32* (0.17)	Yes
High Prestige (Count)	0.12** (0.06)	No
High Prestige (Count)	0.2*** (0.06)	Yes

Note: Robust standard errors clustered around country in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In Table C4 we control for additional covariates which could be related to women's appointment to cabinets: corruption, war, party institutionalization, and party-level quotas. Research generally finds that corruption is negatively associated with advancement of women's political careers (Bjarnegård, 2013; Stockemer & Sundström, 2018), although in the wake of corruption scandals, leaders sometimes advance more women—e.g., appointing them to cabinets or adopting gender quotas—to signal they are cleaning up (Armstrong et al.,

Table C4: Results: Additional Models

Model	Estimate (SE)
Main + Corr + War + Party Ist	2.68** (1.1)
Main + Corr + War + Party Ist + Party Quota	3.19** (1.43)

Note: Robust standard errors clustered around country in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

2022; Valdini, 2019). Additionally, conflict termination can accelerate women’s access to the cabinet (Arriola & Johnson, 2014). Following Hughes and Tripp (2015), we use data from the Uppsala Conflict Data Program to account for whether there was a major armed conflict that resulted in 1,000 fatalities or more during the previous year. We also control for the country’s level of party institutionalization because studies show that when parties are relatively more institutionalized, women are more likely to be included – not least because strong parties are more likely to have clear rules and procedures for selection (Caul 1999; Waylen 2000; Wylie 2018; but see Junqueira & Cunha Silva 2024 for a counterargument). Our index of party institutionalization comes from the Varieties of Democracy dataset. Finally, we control for a conservative measure of party-level quotas. Leftist parties often adopt internal targets before national reforms, and their leaders may be more likely to appoint women (Claveria, 2014). We compiled data on party quotas from several sources (Weeks, 2022; Funk, Hinojosa & Piscopo, 2017; International IDEA, 2023). Although adoption dates are often imprecise, these indicators signal party support for women’s inclusion. To be conservative, we control for whether governing parties had ever adopted a quota.

Two-Way Fixed Effect Models Results

Table C5: Gender Quota Laws and Share of Women in Cabinet

	Dependent variable:	
	% women in cabinet	
	(1)	(2)
Quota law	2.54*** (0.79)	2.19*** (0.78)
Woman leader _{t-1}		0.70 (1.38)
Left executive		2.17*** (0.83)
Women's labor force _{t-1}		0.20*** (0.08)
Stock of polyarchy (95%) _{t-1}		17.20*** (4.56)
Constant	-2.06** (0.81)	-6.61*** (1.51)
Country FE	Yes	Yes
Year FE	Yes	Yes
Observations	5246	5043
R ²	0.68	0.69
Adjusted R ²	0.67	0.68

Note: Robust standard errors clustered around country in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C6: Quota Laws and Share of Women in Cabinets, by Regime Type

	Dependent variable:					
	% women in cabinet					
	Parliamentary		Presidential		Authoritarian	
	(1)	(2)	(3)	(4)	(5)	(6)
Quota law	3.25** (1.47)	3.18** (1.46)	3.08* (1.84)	2.63 (1.79)	1.80 (1.22)	1.62 (1.15)
Woman leader _{t-1}		-0.88 (1.47)		2.71 (2.26)		3.09 (2.65)
Left executive		1.82* (0.94)		3.25** (1.59)		1.43 (1.80)
Women's labor force _{t-1}		-0.01 (0.13)		0.39** (0.17)		0.21* (0.12)
Stock of polyarchy (95%) _{t-1}		5.68 (7.98)		-1.80 (10.87)		26.09*** (9.42)
Constant	4.96*** (1.48)	4.15 (7.53)	0.73 (1.97)	-17.63* (9.18)	0.78 (1.07)	-6.06** (2.37)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1850	1799	1056	1032	2340	2212
R ²	0.71	0.72	0.56	0.58	0.68	0.70
Adjusted R ²	0.70	0.70	0.52	0.55	0.66	0.68

Note: Robust standard errors clustered around country in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C7: Quota Laws and Share of Women in Cabinets, by Prestige of Cabinet Post

	Dependent variable:					
	% women in cabinet					
	Low prestige		Medium prestige		High prestige	
	(1)	(2)	(3)	(4)	(5)	(6)
Quota law	10.33*** (3.57)	9.69*** (3.61)	1.02 (0.99)	0.66 (0.98)	1.57 (1.24)	1.12 (1.29)
Woman leader _{t-1}		-2.65 (3.63)		2.23 (2.21)		-0.56 (1.60)
Left executive		2.42 (2.51)		1.99* (1.08)		1.93* (1.08)
Women's labor force _{t-1}		0.33 (0.22)		0.18* (0.10)		0.13 (0.09)
Stock of polyarchy (95%) _{t-1}		21.05 (17.40)		17.86*** (4.82)		23.22*** (8.11)
Constant	3.48 (3.23)	1.63 (4.85)	-2.31** (0.95)	-7.53*** (1.64)	-6.30*** (1.00)	-9.30*** (2.14)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4129	4004	5239	5039	5244	5041
R ²	0.39	0.39	0.60	0.61	0.34	0.35
Adjusted R ²	0.36	0.36	0.58	0.60	0.31	0.32

Note: Robust standard errors clustered around country in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C8: Quota Laws and Share of Women in Cabinets, by Quota Shock

	Dependent variable:			
	% women in cabinet			
	High shock		Low shock	
	(1)	(2)	(3)	(4)
Quota law	2.78*** (1.07)	2.51** (1.04)	1.86** (0.84)	1.38* (0.81)
Woman leader _{t-1}		0.62 (1.48)		0.73 (1.53)
Left executive		1.72** (0.81)		1.60* (0.92)
Women's labor force _{t-1}		0.21*** (0.07)		0.19** (0.08)
Stock of polyarchy (95%) _{t-1}		20.66*** (4.94)		19.35*** (4.77)
Constant	-2.37*** (0.91)	-7.31*** (1.44)	-1.37 (0.85)	-5.86*** (1.65)
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	4740	4539	4786	4584
R ²	0.69	0.71	0.69	0.71
Adjusted R ²	0.68	0.69	0.68	0.69

Note: Robust standard errors clustered around country in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C9: Quota Laws and Cabinet Composition: Combined Count Models

	Dependent variable:											
	Count of women / total cabinet counts											
	Cabinet size		Number of men		Number of women		Low prestige		Medium prestige		High prestige	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Quota law	0.04 (0.02)	0.03 (0.02)	0.01 (0.02)	0.01 (0.02)	0.20*** (0.05)	0.20*** (0.05)	0.31*** (0.12)	0.31*** (0.12)	0.17*** (0.06)	0.16*** (0.06)	0.24 (0.15)	0.23 (0.15)
Woman leader _{t-1}		-0.02 (0.03)		-0.04 (0.05)		-0.01 (0.06)		-0.02 (0.11)		-0.02 (0.08)		-0.02 (0.13)
Left executive		0.01 (0.02)		-0.01 (0.02)		0.14*** (0.04)		0.06 (0.08)		0.12** (0.05)		0.28*** (0.09)
Women's labor force _{t-1}		0.00 (0.00)		-0.00 (0.00)		0.01 (0.01)		0.02*** (0.01)		0.00 (0.01)		0.01 (0.01)
Stock of polyarchy (95%) _{t-1}		-0.09 (0.14)		-0.32** (0.14)		0.67* (0.37)		0.01 (0.91)		0.79** (0.39)		1.41 (0.95)
Constant	3.09*** (0.02)	3.05*** (0.04)	3.12*** (0.02)	3.14*** (0.04)	-0.38*** (0.09)	-0.58*** (0.12)	-1.43*** (0.19)	-1.63*** (0.24)	-0.74*** (0.11)	-0.96*** (0.14)	-36.36*** (1.66)	-25.70*** (1.07)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5259	5049	5259	5049	5259	5049	5246	5043	5246	5043	5246	5043
Pseudo R ²	0.71	0.74	0.72	0.75	0.67	0.67	0.47	0.48	0.62	0.62	0.44	0.44

Note: All specifications are Negative Binomial models with robust standard errors clustered around country (in parentheses).

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D Robustness Checks

Joint Tests of Pretrends

Table D1: Joint Test of Pretrends

Unconditional Parallel Trends Assumption			
Pretrends	OLS estimator	CS (1) estimator	CS (2) estimator
Up to $t - 3$	0.4308	0.6056	0.000
Up to $t - 5$	0.6706	0.7521	0.000
Conditional Parallel Trends Assumption			
Pretrends	OLS estimator	CS (1) estimator	CS (2) estimator
Up to $t - 3$	0.4542	0.7645	0.000
Up to $t - 5$	0.7020	0.9065	0.000

Note: This table reports p-values from joint tests of pretrends up to 3 and 5 periods before the reform, using three estimators (OLS, CS (1) and CS (2)) and two model specifications (unconditional and conditional). CS (1) estimator is the Wald test on the aggregate event-time coefficients, and CS (2) estimator is the joint Wald test across all cohort-specific ATTs for pre-treatment periods.

The CS (2) estimator, which we calculated using the command *estat pretrend* included in the Callaway & Sant’Anna (2021) *drdid* package in Stata, is a Wald test on all underlying cohort-specific ATT estimates for the pre-treatment window. As such, it tests whether every cohort satisfies parallel trends before treatment simultaneously. Since, in a staggered difference-in-differences design like ours, many cohort-specific $ATT(g,t)$ are tested simultaneously, the test is strongly susceptible to (potentially harmless) heterogeneity and noise. This means that *estat pretrend* will reject the null of pre-trends even where there are no actual pre-trends in the data (see, for further details, Callaway & Sant’Anna (2021); Goodman-Bacon (2021); Sun & Abraham (2021)). Hence, we also run a Wald test on the aggregated event-time coefficients, CS (1), which does not reject the null of pre-trends. Taking together the visual event study plots, the OLS estimator and the CS (1) results, we are moderately confident that there are no systematic pre-treatment differences and the evidence supports the validity of the parallel trends assumption underlying our analyses.

Leave-One-Out ATT Estimates

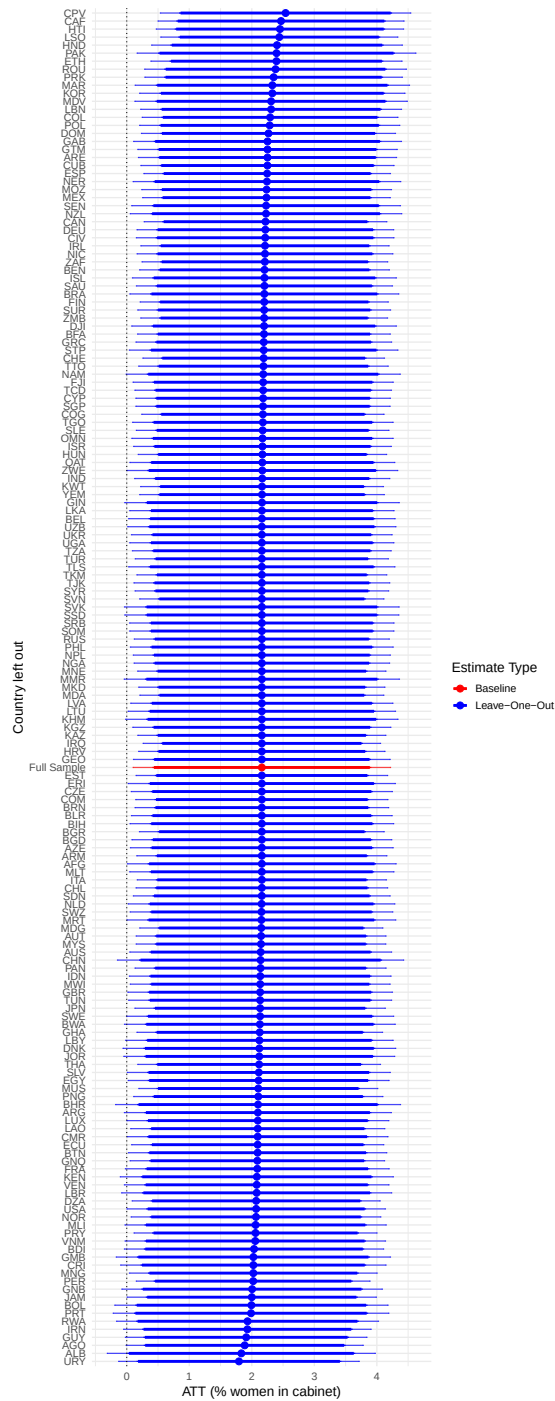


Figure D1: Leave-One-Out Estimates of Quota Effects on Share of Women in Cabinets.

Note: Each point shows the estimated average treatment effect on the treated (ATT) when a country is left out of the sample. The vertical lines represent 90% (thick bar) and 95% (thin bar) confidence intervals. The red point indicates the full-sample (Baseline) estimate.

Figure D.2 reports leave-one-out estimates of the ATT (CS, with controls), obtained by re-estimating the model while sequentially excluding each country. The results show that the estimated effect of quota adoption on women’s cabinet representation remains positive and of similar magnitude throughout. While the precision varies modestly across iterations, the effect consistently retains significance at the 10% level and never changes sign. This stability indicates that the main result is not driven by any single country and is highly robust to sample composition.

Restricting the Sample to Not-Yet-Treated Only

Table D2: Effects of Quota Implementation on Share of Women in Cabinets by Model Type (Not-Yet-Treated Only)

Model	Estimate (SE)	Controls
<i>Overall (H1)</i>		
Full Sample	2.05** (0.99)	No Controls
Full Sample	2.23** (1.05)	Controls
<i>Regime type (H2, H5)</i>		
Parliamentary	3.02*** (0.94)	No Controls
Parliamentary	5.25*** (1.78)	Controls
Presidential	2.7 (3.11)	No Controls
Presidential	3.17 (3.34)	Controls
Authoritarian	1.87 (1.72)	No Controls
Authoritarian	1.99 (1.63)	Controls
<i>Quota shock (H3)</i>		
High Quota Shock	5.2*** (1.11)	No Controls
High Quota Shock	5.02*** (1.2)	Controls
Low Quota Shock	0.1 (1.22)	No Controls
Low Quota Shock	0.48 (1.4)	Controls

Note: Robust standard errors clustered around country in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Covariates include (lagged) democratization (stock of polyarchy (95%)), women’s labor force participation, left executive, and woman leader. For the subsample of authoritarian regimes, the estimates will not compute with the full list of covariates due to not enough control units for any group. For authoritarian regimes, therefore, covariates include democratization and women’s labor force participation only.

Table D3: Effect of Quota Implementation on Share Women in Cabinets, by Prestige of Portfolio (Not-Yet-Treated Only)

Model	Estimate (SE)	Controls
Low Prestige	3.53 (5.58)	No
Low Prestige	0.94 (4.97)	Yes
Medium Prestige	0.55 (1.39)	No
Medium Prestige	0.55 (1.3)	Yes
High Prestige	2.38* (1.4)	No
High Prestige	3.65** (1.55)	Yes

Note: Robust standard errors clustered around country in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D4: Effect of Quota Implementation on Size Cabinet, by Gender, and by Prestige (Not-Yet-Treated Only)

Model	Estimate (SE)	Controls
Cabinet Size	0.84** (0.4)	No
Cabinet Size	0.67* (0.43)	Yes
Number of Men	0.31 (0.42)	No
Number of Men	0.06 (0.51)	Yes
Number of Women	0.53** (0.23)	No
Number of Women	0.64*** (0.22)	Yes
Low Prestige (Count women)	0.25 (0.19)	No
Low Prestige (Count women)	0.04 (0.43)	Yes
Medium Prestige (Count women)	0.31 (0.19)	No
Medium Prestige (Count women)	0.32* (0.17)	Yes
High Prestige (Count women)	0.13** (0.06)	No
High Prestige (Count women)	0.2*** (0.06)	Yes

Note: Robust standard errors clustered around country in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Results by Single High Prestige Portfolio

Table D5: Average Treatment Effects on Women Appointed in High Prestige Portfolios

Model	Estimate (SE)	Controls
Defense	-0.04 (0.05)	No
Defense	0.01 (0.06)	Yes
Foreign Affairs	0.09** (0.04)	No
Foreign Affairs	0.12*** (0.03)	Yes
Interior	0.05 (0.05)	No
Interior	0.11 (0.17)	Yes
Finance	-0.03 (0.04)	No
Finance	0.02 (0.06)	Yes

Note: Dependent variable: woman appointed in the portfolio (binary). Robust standard errors clustered around country in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D6: Quota Laws and Share of Women in Key Portfolios

	Dependent variable:							
	Woman holding portfolio							
	Defense		Foreign affairs		Finance		Interior	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Quota law	0.68 (0.57)	0.62 (0.56)	0.26 (0.30)	0.27 (0.30)	0.17 (0.40)	0.18 (0.39)	0.74* (0.45)	0.79* (0.46)
Woman leader _{t-1}		-0.39 (0.56)		-0.10 (0.37)		0.42 (0.34)		-0.13 (0.38)
Left executive		0.45 (0.29)		0.00 (0.30)		0.07 (0.27)		1.13*** (0.36)
Women's labor force _{t-1}		-0.00 (0.06)		0.01 (0.03)		-0.02 (0.03)		-0.00 (0.06)
Stock of polyarchy (95%) _{t-1}		0.30 (3.92)		1.75 (2.13)		1.52 (2.13)		2.69 (2.60)
Constant	-23.90*** (1.31)	-23.43*** (1.47)	-22.45*** (1.22)	-21.42*** (1.30)	-21.02*** (1.11)	-21.04*** (1.32)	-22.58*** (1.19)	-23.00*** (1.47)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4443	4308	4826	4645	4825	4644	3871	3714
R ²	0.45	0.45	0.35	0.34	0.34	0.34	0.39	0.40

Note: Dependent variable: woman appointed in the portfolio (binary). All specifications are Logit models with robust standard errors clustered around country (in parentheses). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Results Aggregating Low and Medium Prestige Portfolios

Table D7: Average Treatment Effects for Aggregated Low and Medium Prestige Portfolios

Model	Estimate (SE)	Controls
% Women Low–Med Prestige	1.48 (1.21)	No
% Women Low–Med Prestige	1.35 (1.28)	Yes
Low–Med Prestige (Count women)	0.51** (0.21)	No
Low–Med Prestige (Count women)	0.48** (0.19)	Yes

Note: Robust standard errors clustered around country in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D8: Quota Laws and Low–Medium Prestige Portfolios: Share and Count

	Dependent variable:			
	Share of women		Number of women	
	(1)	(2)	(3)	(4)
Quota law	2.40** (0.96)	2.03** (0.94)	0.20*** (0.06)	0.20*** (0.06)
Woman leader _{$t-1$}		1.78 (2.17)		-0.02 (0.07)
Left executive		2.20** (0.97)		0.11*** (0.04)
Women’s labor force _{$t-1$}		0.23** (0.09)		0.01 (0.01)
Stock of polyarchy (95%) _{$t-1$}		17.35*** (5.05)		0.62 (0.39)
Constant	-1.55* (0.94)	-6.76*** (1.67)	-0.33*** (0.09)	-0.55*** (0.13)
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	5239	5039	5246	5043
R ² /Pseudo R ²	0.64	0.66	0.66	0.67

Note: Specifications (1) and (2) are OLS models, and specifications (3) and (4) are Negative Binomial models. All specifications have robust standard errors clustered around country (in parentheses).

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Results Centered on Election Year

Table D9: Average Treatment Effects - Centered on Election Years

Model	Estimate (SE)	Controls
Full Sample	5.83*** (2.05)	No

Note: Robust standard errors clustered around country in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The CS models in Table D9, and in Table D11 in the section below, do not converge when covariates are included. This happens because the Callaway & Sant’Anna (2021) estimator relies on the existence of valid comparison units (never-treated or not-yet-treated) for each treated group–time cell. When including covariates, the estimator discards all observations that have missing values. In the models, the exclusion eliminates several group–time cells or leaves them without any admissible comparison units, which are required for the models to converge. This is not a concern when using the CS estimator, since identification does not require including covariates. Still, it helps reduce residual variance and improve efficiency (thus reducing SEs).

Table D10: Quota Laws and Share of Women in Cabinet: Election Years Only

	Dependent variable:	
	% women in cabinet	
	(1)	(2)
Quota law	2.57** (1.20)	2.36* (1.20)
Woman leader _{t-1}		0.96 (1.74)
Left executive		1.31 (1.05)
Women's labor force _{t-1}		0.12 (0.11)
Stock of polyarchy (95%) _{t-1}		11.26* (6.68)
Constant	-0.60 (2.21)	-4.98* (3.03)
Country FE	Yes	Yes
Year FE	Yes	Yes
Observations	1102	1070
R ²	0.70	0.71
Adjusted R ²	0.63	0.64

Note: Robust standard errors clustered around country in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Results by Regime Type and Shock

Tables D11 and D12 report CS and TWFE estimates by regime type and quota shock. Because these models rely on smaller subsets of countries, they are less precisely estimated than the full-sample results and should be interpreted with caution. Note also that the models for Authoritarian regimes (both low- and high-shock), and for high-shock Parliamentary regimes, do not converge when covariates are included, because of missing data issues, as discussed in the section above.

Table D11: Average Treatment Effects

Model	Estimate (SE)	Controls
High Quota Shock Presidential	4.03 (3.73)	No
High Quota Shock Presidential	6.53 (4.41)	Yes
Low Quota Shock Presidential	1.75 (3.02)	No
Low Quota Shock Presidential	1.45 (5.31)	Yes
High Quota Shock Authoritarian	7.19*** (1.51)	No
Low Quota Shock Authoritarian	0.23 (1.18)	No
High Quota Shock Parliamentary	3.87*** (1.12)	No
Low Quota Shock Parliamentary	2.32** (0.89)	No
Low Quota Shock Parliamentary	2.43 (3.46)	Yes

Note: Robust standard errors clustered around country in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Still, a clear pattern emerges in Tables D11 and D12: high-shock quotas generate the largest and most robust increases in women’s cabinet representation –positive overall (see Figure 3 in main) and significant within both parliamentary and authoritarian systems, though not in presidential regimes. Low-shock quotas are not significant overall and only reach significance in parliamentary democracies, with smaller effects. These findings indicate that quota impacts hinge more on the size of the parliamentary shock than on regime type alone, aligning with a supply-side mechanism in which substantial increases in women’s parliamentary representation expand the pool of eligible “ministrables.”

Table D12: Gender Quota Laws, Regime Type, Shock Size, and Share of Women in Cabinet

	Dependent variable:											
	% women in cabinet											
	Presidential, HS		Presidential, LS		Authoritarian, HS		Authoritarian, LS		Parliamentary, HS		Parliamentary, LS	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Quota law	4.54 (3.17)	3.11 (3.11)	2.43 (1.92)	1.59 (1.85)	5.11* (2.67)	4.57* (2.58)	0.69 (1.10)	0.23 (1.06)	4.50* (2.42)	4.14* (2.46)	3.67** (1.78)	3.66* (1.88)
Woman leader _{t-1}		4.74** (2.34)		2.79 (3.16)		-4.00 (2.97)		3.71 (2.80)		-1.47 (1.51)		-1.67 (1.48)
Left executive		2.28 (1.57)		2.08 (2.45)		-0.17 (2.75)		0.29 (2.53)		2.53** (1.10)		1.71* (1.04)
Women's labor force _{t-1}		0.37** (0.15)		0.35 (0.23)		0.17 (0.14)		0.18* (0.10)		0.19* (0.10)		-0.02 (0.14)
Stock of polyarchy (95%) _{t-1}		-0.39 (11.24)		10.82 (16.83)		34.95*** (10.60)		28.45*** (9.88)		5.90 (9.31)		3.81 (8.70)
Constant	-1.07 (2.75)	-19.51** (8.09)	7.56*** (2.63)	-16.16 (14.02)	8.60*** (1.68)	-5.79 (11.21)	1.77* (0.99)	-4.34* (2.38)	4.49*** (1.63)	-6.22 (5.94)	-7.79*** (1.79)	-6.70 (9.03)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	614	600	689	674	1475	1388	1828	1725	1295	1259	1541	1496
R ²	0.61	0.64	0.49	0.52	0.69	0.73	0.66	0.69	0.75	0.75	0.73	0.74
Adjusted R ²	0.58	0.60	0.44	0.47	0.67	0.71	0.64	0.67	0.73	0.74	0.72	0.72

Note: Robust standard errors clustered around country in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

References

- Armstrong, Brenna, Tiffany D Barnes, Diana Z O'Brien & Michelle M Taylor-Robinson. 2022. "Corruption, accountability, and women's access to power." *The Journal of Politics* 84(2):1207–1213.
- Arriola, Leonardo R & Martha C Johnson. 2014. "Ethnic politics and women's empowerment in Africa." *American Journal of Political Science* 58(2):495–510.
- Bjarnegård, Elin. 2013. *Gender, informal institutions and political recruitment: Explaining male dominance in parliamentary representation*. Springer.
- Callaway, Brantly & Pedro HC Sant'Anna. 2021. "Difference-in-differences with multiple time periods." *Journal of econometrics* 225(2):200–230.
- Caul, Miki. 1999. "Women's representation in parliament: The role of political parties." *Party politics* 5(1):79–98.
- Claveria, Silvia. 2014. "Still a 'male business'? Explaining women's presence in executive office." *West European Politics* 37(5):1156–1176.
- Funk, Kendall D, Magda Hinojosa & Jennifer M Piscopo. 2017. "Still left behind: gender, political parties, and Latin America's pink tide." *Social Politics* 24(4):399–424.
- Goodman-Bacon, Andrew. 2021. "Difference-in-differences with variation in treatment timing." *Journal of econometrics* 225(2):254–277.
- Hughes, Melanie M & Aili Mari Tripp. 2015. "Civil war and trajectories of change in women's political representation in Africa, 1985–2010." *Social forces* 93(4):1513–1540.
- International IDEA, Stockholm. 2023. "Gender Quotas Database."
- Junqueira, Andrea & Patrick Cunha Silva. 2024. "Strengthening the Party, Weakening the

- Women: Unforeseen Consequences of Strengthening Institutions.” *The Journal of Politics* 86(2):000–000.
- Stockemer, Daniel & Aksel Sundström. 2018. “Age representation in parliaments: Can institutions pave the way for the young?” *European Political Science Review* 10(3):467–490.
- Sun, Liyang & Sarah Abraham. 2021. “Estimating dynamic treatment effects in event studies with heterogeneous treatment effects.” *Journal of econometrics* 225(2):175–199.
- Valdini, Melody E. 2019. *The inclusion calculation: Why men appropriate women’s representation*. Oxford University Press.
- Waylen, Georgina. 2000. “Gender and democratic politics: A comparative analysis of consolidation in Argentina and Chile.” *Journal of Latin American Studies* 32(3):765–793.
- Weeks, Ana Catalano. 2022. *Making Gender Salient: From Gender Quota Laws to Policy*. Cambridge University Press.
- Wylie, Kristin N. 2018. *Party institutionalization and women’s representation in democratic Brazil*. Cambridge University Press.